Microservice-Based Neural Network Inference Across Architectural Splits, Cloud Deployment, and Security Hardening

A Staged Empirical Evaluation of ResNet-18 on Azure Kubernetes Service

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Abstract

Neural network inference is increasingly deployed inside cloud-native software systems that must balance latency, deployability, scalability, and isolation. Splitting a model across microservices can improve modularity and make security boundaries more explicit, but it also forces intermediate activations to cross service interfaces. Each boundary adds serialization, transport, scheduling, and coordination cost. The central problem studied in this thesis is therefore not only where to split a neural network, but whether a decomposition that appears acceptable in a controlled local setting remains credible once it is moved into Kubernetes, Azure, and security-hardened deployment contexts.

This thesis addresses that problem through a staged experimental evaluation of ResNet-18 inference. A proof-of-concept gRPC-based microservice framework is used to compare monolithic inference, two-part model partitions, and synchronously chained multi-service deployments. The evaluation first screens coarse architectural split points under controlled local execution, then refines the selected region and explains the observed behaviour using activation-transfer size and compute distribution. The same decomposition family is then evaluated in local Kubernetes and Azure Kubernetes Service, after which the selected Azure deployment is hardened with service identity and mutual TLS. Confidential-execution mechanisms are treated as a later incremental protection layer on top of the secured Azure path.

The results show that monolithic execution remains the fastest baseline, but that not all microservice decompositions are equally costly. Early splits suffer from large activation transfers, while very late splits can become compute-degenerate. Mid-network boundaries provide the most credible two-part trade-off. Fine-grained refinement within this region does not materially improve latency, supporting the use of coarse architectural boundaries for later stages. In Kubernetes and Azure, latency increases with service count, but the increase is bounded and non-linear: later boundaries add less cost when the transferred activations are smaller. Azure preserves the condition ordering observed in local Kubernetes, indicating directional transfer despite different absolute latencies. Adding mutual TLS and service identity introduces measurable but bounded overhead, alongside clear operational complexity from sidecars, policies, and readiness delays.

Overall, the thesis concludes that microservice-based inference is viable only when architectural partitioning, deployment context, and security techniques are evaluated

together. The main engineering trade-off is not between monolithic and distributed inference in isolation, but between deployability and stronger isolation on one side and cumulative boundary-related overhead on the other.

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Chapter 1

Introduction

1.1 Context

Deep neural network inference is increasingly embedded in software systems that must satisfy not only predictive requirements, but also operational ones such as latency, deployability, maintainability, scalability, and isolation. Edge-intelligence and split-computing surveys describe this shift as part of a broader move from purely centralized inference toward device, edge, and cloud collaboration [23, 35]. In many practical deployments, inference still runs as a monolithic component inside a single process. That architecture minimizes boundary-crossing overhead, but it also couples model execution into one deployment unit and limits architectural flexibility. A microservice-oriented design offers a different trade-off: the inference pipeline can be decomposed across service boundaries, allowing clearer separation of responsibilities, independent deployment, and more explicit control over resource placement.

For deep learning systems, however, decomposition is not only a software architecture decision. It is also a systems problem shaped by the structure of the neural network itself and by the environment in which the resulting services run. When a model is split across an execution boundary, intermediate activations must be serialized, transferred, and reconstructed before downstream computation can continue. Prior work on split computing and distributed DNN inference shows that these costs are highly sensitive to where the boundary is placed. DNNSplit demonstrates that split points should be selected systematically with respect to latency and cost rather than by arbitrary layer choice [18]. DeeperThings and survey work on DNN partitioning likewise show that early layers often produce large feature maps that are expensive to transmit, while later boundaries may reduce transfer volume but shift the compute balance too far toward one side of the partition [31, 33].

The deployment environment introduces an additional dimension to this problem. A split that is acceptable in a controlled local benchmark may behave differently once it is moved into containerized, orchestrated, or cloud-hosted execution, because container platforms, managed Kubernetes services, and cloud infrastructure add their own scheduling, networking, and virtualization effects [8, 9, 16]. Beyond performance, there is also a security dimension. As inference services move into shared or remote infrastructure,

communication-level protections such as service identity and mutual TLS become relevant because they strengthen inter-service trust boundaries, while service-mesh studies show that this protection can add measurable runtime and resource cost [5, 36]. Mutual TLS terminates at the pod boundary, however, so intermediate activations are still exposed to the runtime environment of the downstream service; prior work shows that an untrusted remote partition holding only those activations can reconstruct the input [12], motivating an additional layer of protection. Confidential-computing techniques such as trusted execution environments may add such a layer, but their overhead and protection boundary are workload- and platform-dependent [10, 25], and recent work shows that confidential VMs still have residual attack surfaces beyond memory encryption, including malicious-host interrupt injection [29]. This makes microservice-based inference a broader software engineering problem involving architectural partitioning, deployment infrastructure, and security mechanisms rather than only local latency optimisation.

1.2 Problem Statement

The central problem addressed in this thesis is how to design and evaluate deep neural network inference as a microservice system without losing control over performance. A poor split can make a distributed design clearly inferior to monolithic execution. Prior partitioning systems show that if the boundary is placed too early, the system may incur excessive latency because large activation tensors must cross the service interface; if the boundary is placed too late, communication cost decreases, but the downstream service may be left with too little computation to justify the split [15, 18, 31]. In other words, minimizing activation size alone is not enough; the chosen boundary must also preserve a meaningful compute distribution across the resulting services.

That architectural problem is only the first layer of the thesis. After a defensible split region has been identified locally, the same decomposition logic must also be evaluated under progressively more realistic deployment conditions. Container orchestration and cloud deployment introduce additional scheduling, networking, and platform overhead that may alter the trade-offs seen in a local benchmark [6, 8]. Finally, if the selected deployment path is security-hardened, the system may incur further overhead in exchange for stronger inter-service authentication, encrypted communication, and confidential execution for selected services [34, 36]. The thesis therefore treats the problem as a staged evaluation pipeline: first identify and explain suitable boundaries under controlled local conditions, then study how a coarse-stage service-chain family behaves in Kubernetes and Azure deployments, and finally measure the impact of adding communication-level and VM-level security hardening to the selected Azure deployment path.

1.3 Research Questions

This thesis is organised around two broader research strands. RQ1 studies how architectural split choice and deployment context affect microservice-based inference. RQ2 studies the performance, operational, and protection-scope trade-offs introduced when the selected Azure deployment path is security-hardened. The research questions are:

RQ1.1 Which coarse architectural stage boundaries provide the most suitable two-part microservice decompositions of ResNet-18 under controlled local execution, and how do their latency and boundary-crossing costs compare to monolithic inference?

RQ1.2 How does finer-grained refinement within the late-stage coarse region carried forward from RQ1.1 affect latency and communication-related overhead for two-part microservice inference?

RQ1.3 How do activation-transfer size and compute distribution explain the latency and boundary-crossing behaviour observed for the decomposition choices evaluated in RQ1.1 and RQ1.2?

RQ1.4 How does increasing the number of microservices affect end-to-end inference cost when ResNet-18 is decomposed into synchronously chained coarse architectural stages under in-cluster Kubernetes execution?

RQ1.5 Does the local Kubernetes latency pattern observed for the coarse-stage ResNet-18 microservice chain transfer to Azure Kubernetes Service, and how do the absolute and relative costs change under managed cloud execution?

RQ1.5b When every chain hop is forced across a node boundary, how much additional cost is introduced relative to the same-node RQ1.5 baseline?

RQ2.1 What performance and operational overhead is introduced when the selected AKS microservice inference path is hardened with service identity, mutual TLS, and inter-service authorization policy?

RQ2.2 What additional performance and operational overhead is introduced when VM-level confidential execution is added to the already communication-secured AKS inference chain, when the downstream protected service is placed on AMD SEV-SNP confidential nodes?

RQ2.3 Which security-hardening configuration provides the most appropriate trade-off between performance cost, operational complexity, and protection scope for the selected AKS microservice inference deployment?

RQ1.1 and RQ1.2 form the local split-selection stages. RQ1.3 is an explanatory synthesis that uses activation-transfer size and internal compute distribution to interpret those local results. RQ1.4 and RQ1.5 then evaluate the same coarse-stage chain family in local Kubernetes and AKS, while RQ1.5b acts as a sensitivity stage for cross-node placement. RQ2.1 adds communication-level hardening through service identity, mutual TLS, and authorization policy. RQ2.2 adds selective AMD SEV-SNP VM-level confidential execution for the downstream protected service. RQ2.3 synthesises those security stages against the plain AKS baseline.

1.4 Approach

The thesis uses a staged experimental methodology centred on ResNet-18, a residual architecture whose stage structure provides natural coarse split candidates [11]. The initial benchmark compares one monolithic baseline against four two-part split configurations placed after `layer1`, `layer2`, `layer3`, and `layer4`. These boundaries correspond to major architectural stages in the network and therefore provide a principled coarse sweep from early to late splits. This first stage is executed locally with CPU-only inference, FP32

precision, gRPC over localhost, fixed warmup and measurement windows, and functional parity checks between monolithic and split execution so that boundary effects can be isolated under controlled conditions, following systems-benchmarking guidance that emphasises controlled comparisons and explicit reporting of measurement context [13].

The local selection stages are then refined and explained before the deployment context is changed. A block-level split inside the carried-forward region tests whether finer granularity changes the conclusion, while an internal timing study of the monolithic model supports the interpretation of activation-transfer and compute-distribution effects. The same model and benchmark contract are then carried into a predefined Kubernetes service-chain family, from a one-service monolithic deployment to a five-service coarse-stage chain, and this family is evaluated first in local Kubernetes and then on Azure Kubernetes Service.

The final stages add security hardening to the selected Azure path. RQ2.1 compares the plain AKS chain against a managed-Istio configuration with service identity, mutual TLS, and inter-service authorization policy. RQ2.2 keeps that communication-security contract fixed and moves the downstream protected service onto AMD SEV-SNP confidential nodes to measure the incremental cost of VM-level confidential execution. RQ2.3 then compares the plain, communication-secured, and selectively confidential configurations. This design follows the broader literature showing that DNN partitioning must be evaluated as a trade-off between communication and computation rather than as a purely architectural choice [19, 26, 28]. In this thesis, that trade-off is examined across multiple deployment and protection layers rather than being treated as a purely local benchmarking problem.

1.5 Outline

The remainder of this thesis is organised as follows. Chapter 2 introduces the technical background required for understanding deep neural network inference, ResNet-style architectures, microservice-based deployment, and the broader systems context in which distributed inference is evaluated. Chapter 3 reviews prior literature on split computing, DNN partitioning, distributed inference, and the trade-offs that motivate this thesis. Chapter 4 describes the research methodology and the staged evaluation design used to move from local boundary screening to broader deployment and security contexts. Chapter 5 presents the empirical evaluation stages, beginning with RQ1.1 and continuing through the later deployment-oriented and security-hardening analyses. Chapter 6 interprets those findings in relation to the research questions and discusses their implications and limitations. Finally, Chapter 7 summarises the thesis and outlines directions for future work.

Chapter 2

Background

This chapter introduces the technical concepts needed to interpret the staged experiments. It explains the ResNet-18 architecture used as the inference workload, the meaning of a split boundary in distributed DNN inference, the difference between coarse and block-level split granularity, and the service-chain execution model used in the Kubernetes and AKS stages. It also introduces the protection boundaries used in the security-hardening stages. The chapter refers to literature where it clarifies a technique or concept; the more comparative discussion of prior work appears in chapter 3.

2.1 ResNet-18 as the Inference Workload

ResNet-18 can be viewed as three major parts for the purposes of this thesis: an initial stem, four residual stages, and a classification head. The stem performs the first feature extraction and spatial downsampling through convolution, normalisation, activation, and max pooling. The four residual stages, denoted `layer1` through `layer4`, contain the main residual-block computation of the network. The classification head then reduces the final feature map through average pooling and applies the final fully connected classifier. This stage-based view follows the residual-network design introduced by He et al., where successive stages reduce spatial resolution while increasing channel depth [11].

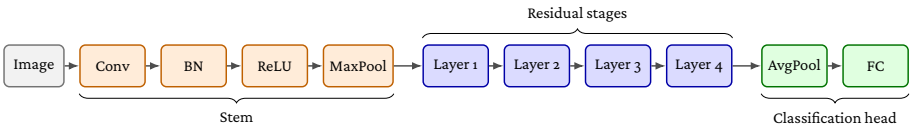


Figure 2.1: Top-level ResNet-18 inference pipeline used in this thesis. The model is treated as a stem, four residual stages, and a classification head.

2.2 Split Inference and Partition Boundaries

In split inference, a neural network forward pass is divided into execution segments. An upstream segment computes an intermediate activation tensor, transfers that tensor to another execution environment, and a downstream segment completes the forward pass. Prior split-inference systems use this same basic abstraction when partitioning neural network inference across devices, edge nodes, or cloud resources [7, 15].

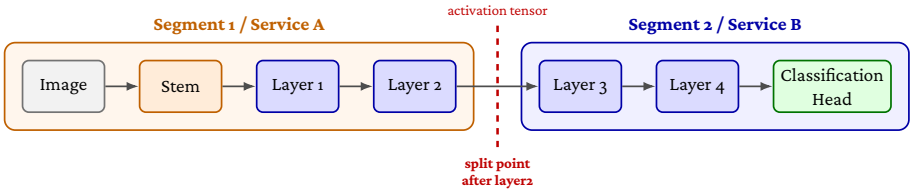


Figure 2.2: Example two-segment decomposition produced by a split after `layer2`. The first segment executes the stem through `layer2`; the second segment receives the intermediate activation tensor and executes `layer3` through the classification head.

A split boundary is therefore both an architectural position in the model and a runtime interface between two execution segments. For example, `split_after_layer2` creates the two-segment decomposition shown in fig. 2.2: the first segment executes the model up to the boundary, and the second segment resumes computation from the transferred activation tensor.

2.3 Coarse Split Points in ResNet-18

The stage structure of ResNet-18 motivates the coarse split choices used later in RQ1.1. The thesis does not treat every primitive operation as an equally meaningful split candidate. Instead, the initial coarse sweep uses the four residual-stage boundaries after `layer1`, `layer2`, `layer3`, and `layer4`. These boundaries preserve the architectural stage structure of ResNet-18 and provide a controlled progression from early, high-activation regions to later, lower-activation regions.

Splits inside the stem are excluded from the primary coarse sweep because they would place boundaries between tightly coupled primitive operations before substantial activation-size reduction. Splits inside the classification head are also excluded because the head contains only the final pooling and linear classification operations, leaving one side of the partition with negligible computation. This framing follows partitioning literature that treats split selection as a trade-off among activation-transfer cost, compute placement, and deployment context rather than as a search over depth alone [18, 31, 33].

As shown in fig. 2.3, RQ1.1 evaluates split points after the four major residual stages rather than inside the stem or classification head.

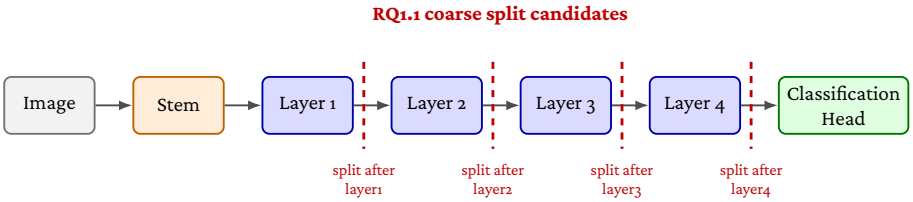


Figure 2.3: Coarse ResNet-18 partition boundaries evaluated in RQ1.1. The stem and classification head are collapsed into single blocks, while the four residual stages are shown as the candidate split locations.

2.4 Block-Level Structure and Fine-Grained Splits

Each residual stage in ResNet-18 contains two BasicBlock containers, and each BasicBlock contains two convolution operations. The first block of layers 2–4 also uses a projection shortcut for downsampling. This internal structure is hidden in the coarse stage-level view, but it matters when reasoning about finer split boundaries [11].

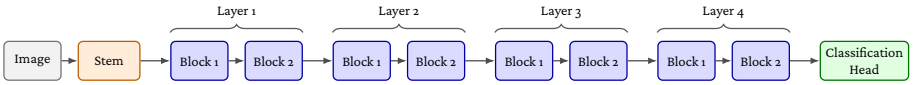


Figure 2.4: Simplified ResNet-18 block structure. The stem and classification head are shown as single components, while each residual stage is shown as two basic blocks.

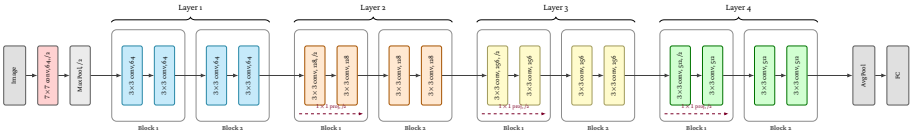


Figure 2.5: Detailed ResNet-18 structure with BasicBlock containers. Each residual stage contains two BasicBlocks, and each BasicBlock contains two convolution operations. The first block of layers 2–4 includes a projection shortcut for downsampling.

The block-level view matters because two boundaries with the same activation tensor can still assign different amounts of computation to the upstream and downstream services. The refined boundary inside `layer3` used later in the thesis should therefore be read as a compute-placement distinction, not only as an activation-size distinction. Prior work on fine-grained and hierarchical DNN partitioning similarly argues that useful split selection requires awareness of the model's internal computational structure [20, 28, 32].

2.5 Service Boundaries and RPC Cost

A split boundary becomes a service boundary when the execution segments are packaged as independently addressable services. In that setting, the boundary-crossing interval

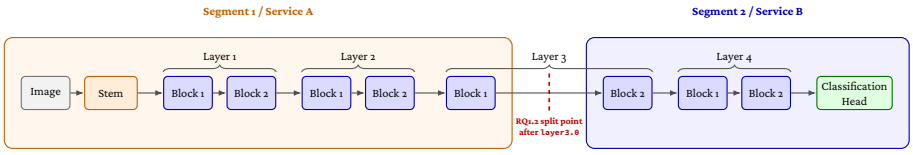


Figure 2.6: Block-level two-segment decomposition used in RQ1.2. The refined split point is placed inside `layer3`, after the first residual block. Segment 1 executes the stem through `layer3.0`; Segment 2 receives the intermediate activation tensor and executes `layer3.1` through the classification head.

used later in the thesis should be read as a compound systems cost rather than as network transfer alone. It includes tensor serialization, RPC handling, transport through the service interface, downstream deserialization, remote computation after the boundary, and response handling.

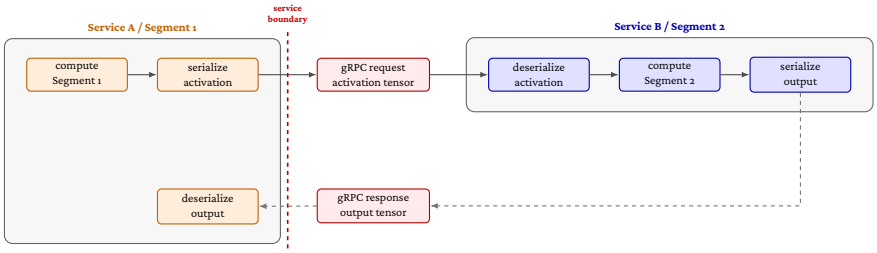


Figure 2.7: Runtime cost components introduced by a two-service inference boundary. In addition to the model computation on each side of the split, the intermediate activation must be serialized, transferred through gRPC, deserialized by the downstream service, and followed by response serialization and reconstruction. The boundary-crossing interval therefore includes serialization, RPC transport, remote execution, and response reconstruction.

Recent microservice systems work shows that RPC paths can accumulate overhead through protocol processing, memory copies, socket operations, and network-stack traversal, even before the application-specific computation is considered [2, 37].

2.6 Kubernetes Service Chains

In the Kubernetes and AKS stages, each model segment is packaged as a separate service and communicates through synchronous gRPC forwarding. This makes the inference pipeline closer to a cloud-native microservice deployment, but it also introduces platform effects from container execution, service networking, CNI behavior, and managed-cluster infrastructure. The single-node and multi-node placements used later are shown in figs. 2.8 and 2.9.

Prior Kubernetes and cloud-performance studies show that container execution, service networking, CNI behavior, and managed infrastructure can affect measured

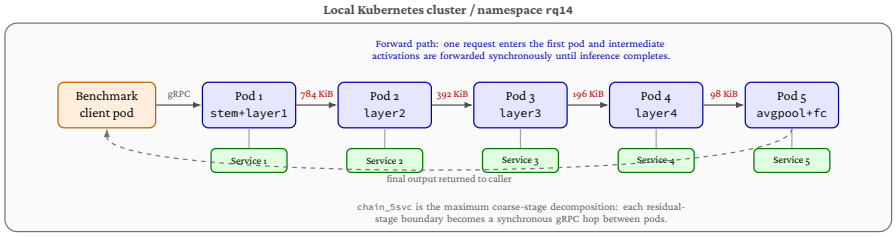


Figure 2.8: Kubernetes pod and service placement for the RQ1.4 five-service chain. The benchmark client and all inference pods are placed on the same Kubernetes node, while each model segment is exposed through a Kubernetes Service and communicates using synchronous gRPC forwarding. The labels above the arrows show the activation-transfer size at each coarse ResNet-18 boundary.

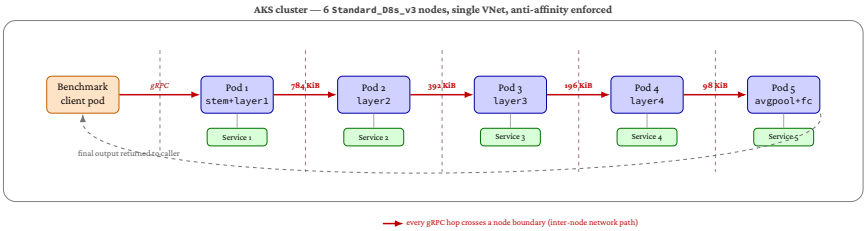


Figure 2.9: RQ1.5b AKS multi-node placement of `chain_5svc`. Each chain segment lands on a distinct cluster node; the benchmark client pod lands on a sixth dedicated node. Every gRPC hop therefore crosses a node boundary, in contrast to the same-node layout used for RQ1.4 and RQ1.5 (fig. 2.8).

latency and variability. This is why the terminology in this thesis distinguishes local process benchmarks, local Kubernetes benchmarks, and managed AKS benchmarks as separate execution contexts [6, 8, 16].

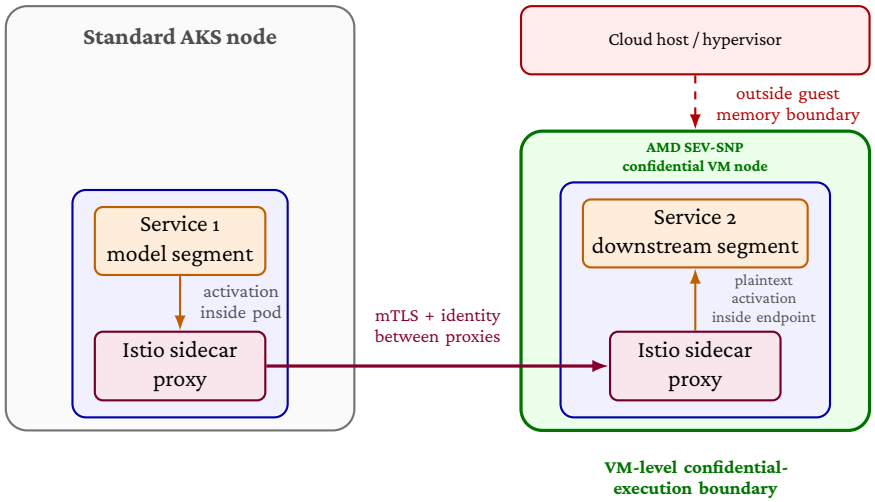
2.7 Communication and Runtime Protection Boundaries

The security-hardening stages use two distinct protection layers. The first layer is communication-level hardening. In a service-mesh deployment, sidecar proxies can attach identity to workloads, establish mutually authenticated TLS channels between services, and enforce authorization policy based on those identities. This strengthens the service-to-service path because a downstream service can reject traffic that does not come from the expected peer identity, while the traffic between proxies is encrypted and mutually authenticated [4, 5].

This protection boundary is still a communication boundary. Once a request has reached the downstream endpoint and the activation tensor has been deserialized, the plaintext activation exists inside the receiving service's runtime environment. Communication hardening therefore protects the inter-service path, but it does not by itself protect the downstream service from the infrastructure on which it runs or from software

inside the receiving runtime.

The second layer used later in the thesis is VM-level confidential execution. On Azure, confidential VM nodes backed by AMD SEV-SNP protect the guest VM's memory and execution state against parts of the host platform and hypervisor, shifting the relevant boundary from the network path to the VM boundary [3]. In this thesis, this mechanism is applied selectively to the downstream service in the secured two-service chain. The term *VM-level confidential execution* is used deliberately: the protected unit is the VM that hosts the Kubernetes node, not an application-level enclave around only the model code or activation tensor.



mTLS protects traffic on the service-to-service path. SEV-SNP shifts the downstream service into a confidential VM boundary, but plaintext still exists inside the trusted endpoint after decryption.

Figure 2.10: Protection boundaries in the secured two-service AKS deployment. The service mesh provides communication-level protection between sidecar proxies using mTLS and workload identity. AMD SEV-SNP confidential nodes add a VM-level host-protection boundary around the downstream service's Kubernetes node. The figure is schematic: plaintext activations still exist inside the endpoint after mTLS termination, and the VM-level boundary does not remove all trusted software or residual attack surfaces.

This distinction, shown in fig. 2.10, is important for interpreting the security results. Service-mesh mTLS and authorization answer whether the selected AKS inference path can be protected at the communication level and what that costs. SEV-SNP-backed confidential nodes answer a different question: what extra cost is paid when the downstream service is also placed inside a VM-level host-protection boundary. Neither layer removes every trusted component or eliminates all residual attack surfaces; confidential-VM systems still have bounded threat models and known residual risks [10, 29].

Chapter 3

Related Work

This chapter positions the thesis within prior work on split neural-network inference, cloud-native deployment overhead, and security hardening for microservice systems. The central distinction is that most prior partitioning work focuses on selecting or optimising split points, while this thesis follows one workload through a staged software-engineering evaluation: local split screening, Kubernetes deployment, managed AKS transfer validation, and security hardening.

3.1 Split Computing and DNN Partitioning

Early split-computing systems established that neural-network inference can be divided between execution environments, but that the split point determines whether the distributed design is useful. Neurosurgeon models the latency trade-off between mobile and cloud execution and shows that different layers have different computation and data transfer profiles [15]. JointDNN frames partitioning as an optimisation problem in which upload and download costs depend on the intermediate tensor at the chosen boundary [7]. DeeperThings extends this line of work to fully distributed CNN inference on constrained edge devices and reinforces the observation that early feature maps can be expensive to move because they remain spatially large [31].

More recent work broadens the partitioning objective beyond a single latency value. DNNSplit selects split points by jointly considering latency and cost in multi-tier settings [18]. Genetic-algorithm and delay-aware approaches likewise treat the split point as dependent on device capability, bandwidth, throughput, and parallelism constraints [19, 26]. Survey work on DNN partitioning and split computing summarizes this literature as a multi-objective problem involving activation size, compute allocation, deployment tier, and runtime conditions [33]. This thesis adopts that view but keeps the optimisation scope deliberately narrow: it studies a fixed ResNet-18 workload and evaluates how a small number of architecturally meaningful split choices behave across deployment layers.

3.2 Granularity and Internal Model Structure

Several studies show that coarse layer depth is not enough to explain split performance. Fine-grained elastic partitioning demonstrates that more detailed placement decisions can improve distributed DNN inference when the placement is aware of both computation and communication conditions [28]. Hierarchical partitioning similarly combines global partitioning with local refinement, showing that staged decisions can account for heterogeneity at both cluster and node levels [32]. Pareto-front analysis of DNN partitioning further shows that execution time can vary across blocks inside the same model, so boundaries with similar activation sizes may still differ in compute distribution [20]. BottleFit and BottleNet++ approach the problem from the feature-compression side, showing that intermediate representations can be compressed or shaped to reduce communication cost, but that the position of the boundary remains central to the latency and accuracy trade-off [21, 30].

The experiments in this thesis use these findings to justify a coarse-to-fine design. RQ1.1 evaluates residual-stage boundaries in ResNet-18, and RQ1.2 adds a block-level boundary inside the carried-forward region. The goal is not to claim that the selected boundaries are globally optimal for all DNNs. Instead, the goal is to show how activation-transfer size and compute placement interact in a reproducible microservice setting.

3.3 Kubernetes and Cloud-Native Inference

Moving from local split inference to Kubernetes changes the systems context. A microservice boundary is not only a model partition; it is also an RPC path with serialization, protocol handling, socket operations, scheduling effects, and service networking. Recent RPC and microservice studies show that communication overhead can come from layered protocol processing and network-stack traversal, not only from the application payload itself [2, 37]. Service-mesh work reaches a similar conclusion for proxied microservices: sidecars and data-plane components add workload-dependent overhead through additional critical-path processing [36].

Kubernetes and cloud studies also show why local and managed-cluster results should be compared carefully. Managed Kubernetes performance varies across services and configurations [8]. Container and VM layers can differ from bare-metal execution [9]. CNI plugins and edge-oriented Kubernetes networking can affect communication cost [16]. Cloud environments also exhibit network-performance variability that can change application scalability and measurement stability [6]. For this reason, this thesis does not treat the local Kubernetes and AKS stages as numerical replicas. Instead, RQ1.5 asks whether the ordering and relative overhead trends from local Kubernetes transfer directionally to managed AKS.

3.4 Security Hardening

The security stages build on prior work in service identity, mTLS, service meshes, and confidential computing. Policy-driven Kubernetes security work motivates combining encrypted communication with explicit authorization policy rather than treating TLS

alone as the complete protection mechanism [4]. Service mesh studies show that sidecar-based mTLS can improve service-to-service trust boundaries, but also that it can add latency and CPU/resource overhead, with the cost depending on mesh configuration and workload [5, 36]. Work on service-mesh control-plane trust further cautions that mesh CAs and control planes remain security-critical components [27]. RQ2.1 therefore treats mTLS and authorization as a bounded communication-hardening layer, not as a complete security solution.

Confidential-computing work motivates the second hardening layer. The motivation itself comes from work on activation-level privacy attacks: He et al. [12] demonstrate that a remote partition holding only intermediate feature maps can reconstruct the edge client's input, and that simple noise-based defenses fail. This shows that in-transit mTLS alone does not protect activations once they reach the downstream segment, although their threat model is malicious-remote-service rather than the infrastructure-level adversary that confidential VMs address.

AMD SEV-SNP provides VM-level memory encryption and integrity protection against parts of the host platform [3, 17]. Comparative studies of virtualization-based confidential-computing platforms emphasize that these systems have specific attacker models, attestation mechanisms, and residual limitations [10, 24]. Performance studies show that confidential VM overhead is workload-dependent, with CPU-bound, memory-intensive, and I/O-heavy workloads affected differently [1, 24]. Machine-learning confidential computing surveys identify protected model execution and data privacy as important motivations, while also emphasizing deployment and performance trade-offs [25]. Recent work additionally shows that the hypervisor remains a meaningful adversary even under SEV-SNP: HECKLER [29] uses malicious-interrupt injection to bypass OpenSSH and sudo authentication on production SEV-SNP and TDX, and additionally breaks execution integrity of analytical workloads on SEV-SNP, an attack surface that hardware memory encryption alone does not cover. RQ2.2 follows this bounded framing by measuring VM-level confidential execution on AKS for the downstream service only, rather than claiming application-level enclave isolation or evaluating those residual attack surfaces.

3.5 Positioning of This Thesis

The related literature establishes the individual concerns that motivate this thesis: split points affect activation-transfer and compute costs, microservice and Kubernetes deployment add platform overhead, and security hardening strengthens protection while introducing additional cost. The contribution of this thesis is to connect those concerns in one staged empirical pipeline. Instead of evaluating partitioning, orchestration, and hardening as separate problems, the thesis tracks the same ResNet-18 microservice inference workload from local split selection through Kubernetes, AKS, mTLS/AuthZ, and confidential VM execution.

Chapter 4

Methods

This chapter describes the research design used to answer the research questions defined in Chapter 1. It explains which methods and procedures are used at each stage, why those methods are appropriate, and what the known limitations and pitfalls of the chosen approach are. The thesis addresses a problem that spans software architecture, containerised deployment, cloud infrastructure, and communication-level security. Studying these layers in a single undifferentiated experiment would confound the effects of partition boundary placement, orchestration overhead, and security mechanisms into one mixed result. The methodology therefore follows a deliberate *staged experimental pipeline* in which each stage changes one layer of the system at a time, preserves the model and benchmark contract, and compares against the nearest relevant baseline.

4.1 Overall Research Design

The thesis is organised around two research strands and a staged experimental pipeline. The primary benchmark stages are RQ1.1, RQ1.2, RQ1.4, RQ1.4b, RQ1.5, RQ1.5b, RQ2.1, and RQ2.2. RQ1.3 is an explanatory synthesis supported by a separate monolithic timing experiment, and RQ2.3 is the final security trade-off synthesis. RQ1 studies how architectural split choice and deployment context affect microservice-based neural network inference. RQ2 studies the performance and operational cost of communication-level and runtime-level security hardening applied to the AKS deployment path selected from RQ1.

The stages unfold in the following sequence. RQ1.1 evaluates four coarse ResNet-18 stage boundaries under tightly controlled local CPU-only execution in order to identify which two-part decompositions are credible enough to carry forward before the thesis moves to orchestrated or cloud deployment. RQ1.2 tests one block-level split inside the carried-forward late-stage region to determine whether finer granularity reveals meaningful differences beyond the coarse level; it is a targeted refinement, not a new blind search. RQ1.3 is a hybrid analytical stage that explains the RQ1.1 and RQ1.2 latency patterns using two structural properties of the partition – activation-transfer size and compute distribution – drawing on the prior benchmark summaries and on a separately collected internal timing experiment on the monolithic model, without conducting a

new split benchmark. RQ1.4 moves a predefined family of one-to-five service chains into a local single-node `kind` Kubernetes cluster to measure how end-to-end latency accumulates as synchronous service boundaries are added under in-cluster orchestration. RQ1.5 re-runs the same predefined chain family on Azure Kubernetes Service (AKS) to test whether the local Kubernetes ordering and overhead pattern transfer directionally to managed cloud execution. RQ1.5b is an AKS multi-node sensitivity stage that re-runs the same chain family on a six-node cluster with pod anti-affinity, so every chain hop is forced across a node boundary; it quantifies the inter-node penalty that the single-node RQ1.5 design intentionally hides. RQ2.1 adds managed-Istio mTLS, service accounts, and authorisation policy to the selected AKS deployment path, measuring the resulting latency, resource, and operational overhead using a paired execution design. RQ2.2 adds AMD SEV-SNP VM-level confidential execution to the already communication-secured chain from RQ2.1, measuring the incremental cost of selective downstream-service confidential-node placement (`service2_only`); extending the confidential boundary to both services is out of the thesis-facing scope and is treated as future work. RQ2.3 synthesises the plain AKS baseline from RQ1.5, the mTLS/AuthZ results from RQ2.1, and the selective TEE result from RQ2.2 into a trade-off comparison across cost, operational complexity, and protection scope, without collecting any new benchmark data.

The common design principle across all stages is to change one layer of the system at a time. This is a local design choice for the staged pipeline, motivated by general benchmarking guidance on reproducibility, interpretability, and the careful description of experimental conditions in systems performance work [13], rather than a rule prescribed verbatim by that source.

4.2 System Under Study

4.2.1 Model

The neural network architecture used in all stages is ResNet-18 [11]. The implementation loads the Torchvision default pretrained ImageNet weights and keeps the model fixed across all conditions. ResNet-18 is organised into four residual stages (`layer1` through `layer4`). In this implementation the feature maps after these stages have shapes $64 \times 56 \times 56$, $128 \times 28 \times 28$, $256 \times 14 \times 14$, and $512 \times 7 \times 7$ respectively. Thus, after `layer1`, later stage boundaries halve spatial resolution and double channel count. This structure provides architecturally motivated split candidates with different activation sizes and computational loads.

ResNet-18 is chosen for three reasons. First, its stage structure supports a principled coarse-to-fine screening methodology. Second, its size enables CPU-only inference at latencies realistic enough for measurement while avoiding the confounds of GPU scheduling variance in the early controlled stages. Third, as a standard reference architecture, its behaviour under partition is directly relatable to prior split-computing and DNN-partitioning work [15, 31, 33].

All experiments use FP32 precision with a fixed input shape of $1 \times 3 \times 224 \times 224$ and a batch size of one. Batch size one reflects the latency-oriented framing of the research questions: the thesis measures per-request end-to-end inference latency rather than throughput.

4.2.2 Microservice Framework

A proof-of-concept gRPC-based microservice framework was developed for this thesis. Each service runs as an independent process – or, in the Kubernetes stages, as an independent pod – that receives an input tensor over gRPC, executes its assigned segment of the ResNet-18 model, and transmits the resulting intermediate activation to the next service in the chain. The final service returns the classification output to a benchmark client.

gRPC is chosen as the inter-service protocol because it provides a standard RPC stack over typed Protocol Buffer messages. This keeps the benchmark close to cloud-native microservice practice while still making the transport path explicit; prior microservice-systems work treats RPC communication and Protocol Buffer serialisation as central sources of latency overhead in such systems [37]. The maximum gRPC message size is set to 16 MiB in all stages, which accommodates the largest intermediate activation tensor in the study (approximately 784 KiB for a split after `layer1` in FP32).

The framework supports variable chain depths: the same code path handles two-service chains used in RQ1.1, RQ1.2, and RQ2, and chains of up to five services used in RQ1.4 and RQ1.5. The independent variable in each stage is therefore the choice of split point or the number of chained services, not the communication mechanism.

4.3 Shared Benchmark Contract

To support direct comparison across stages, the thesis defines a shared benchmark contract that is preserved unless a stage explicitly states otherwise. The fixed subject under test is ResNet-18 with pretrained weights, executed on CPU in FP32 with input shape $1 \times 3 \times 224 \times 224$, communicating over gRPC on port 50051 with a maximum message size of 16 MiB.

The shared timing protocol consists of 50 warmup iterations followed by 200 measured iterations per condition execution, with a five-second cooldown between successive conditions, a base random seed of 42 from which a per-round permutation seed seed $+$ round_index is derived so that condition order is independently permuted in each round, and a warmup calibration check using a trailing window of 10 iterations and a coefficient-of-variation (CV) threshold of 0.02. Thesis-facing stages aggregate five rounds or five paired passes, yielding 1,000 measured iterations per condition.

The warmup calibration is implemented as a minimum-plus-stability rule rather than a fixed 50-iteration ceiling. The runner always executes at least the configured 50 warmup iterations, then continues until the trailing-window CV falls below the 0.02 threshold. In the source configs, `max_extra_iterations = -1` denotes no experiment-level extra-warmup cap; the implementation still applies a safety cap of 10 000 additional iterations to prevent non-terminating runs. Each condition's actual `total_iterations` and `extra_iterations_used` are recorded in the frozen `warmup_calibration.json` artefact for that run.

The shared functional validation contract requires local segment validation before timing and live gRPC chain validation for any split or Kubernetes deployment, using an absolute tolerance of atol $= 1 \times 10^{-5}$ and five validation inputs unless a stage explicitly records a stricter tolerance. A condition is not benchmarked unless both checks pass.

In practice, all thesis-facing frozen runs achieve a maximum absolute difference of 0.0 against the monolithic reference, confirming that no numerical drift is introduced by the partition or transport layer; the RQ1.3 internal timing run records the same outcome under the stricter tolerance atol $= 1 \times 10^{-6}$.

This shared contract is the invariant across the staged pipeline: what changes between stages is the infrastructure and security context, not the model, the measurement discipline, or the validation requirements.

4.4 CPU Stabilisation

Systematic CPU stabilisation is applied in all local and Kubernetes stages to reduce variance attributable to hardware noise rather than to the partition boundary. The following controls are requested in every thesis-facing local run. Thread counts for PyTorch intra-op and inter-op threads, and for OpenMP, MKL, and OpenBLAS, are all set to one, ensuring single-threaded model execution. CPU affinity is pinned to one physical core with simultaneous multithreading (SMT) excluded, preventing the benchmark process from migrating between cores or sharing a physical core with a sibling hyperthread. Process scheduling priority is raised using `nice -5` where the runtime permits. The CPU frequency governor is set to `performance` and turbo boost is disabled to prevent frequency scaling from affecting measurements.

These controls are applied symmetrically across all conditions within a benchmark so that differences in measured latency can be attributed to the independent variable rather than to hardware variability.

In containerised and cloud stages, some controls are partially unavailable. In the local `kind` Kubernetes environment (RQ1.4), governor and turbo-disable writes to `/sys` are blocked by the read-only sysfs exposed inside the container; however, the detected governor is already `performance` and turbo is already disabled, so the intended state is achieved through the host configuration. In AKS (RQ1.5 and RQ2), the `nice -5` request is denied due to insufficient privileges and the benchmark runs at `nice = 0`; governor and turbo controls are unavailable through the managed node interface. These constraints are documented per stage and treated as realistic limitations of cloud-native deployment rather than as experimental flaws, because they affect all conditions within a stage symmetrically.

4.5 Measurement and Statistical Analysis

4.5.1 Primary Metrics

End-to-end latency is the primary measurement in all stages, defined as the wall-clock time for one full inference request from the moment the benchmark client dispatches the input to the moment it receives the final classification output.

Results are summarised using the arithmetic mean, median, and standard deviation of per-iteration latencies. The median provides a robust central tendency estimate in the presence of tail spikes. The mean is used as the primary comparison value because the thesis is concerned with expected per-request cost. The 95th-percentile (p95) latency

is reported in the deployment and security stages to characterise tail behaviour under more realistic conditions.

Overhead is reported both in absolute milliseconds and as a percentage of the relevant baseline mean latency. In stages that compare conditions across environments (RQ1.5), within-environment normalised overhead is the primary comparison metric rather than raw absolute latency, because absolute values are environment-specific and do not transfer across different host CPUs, kernels, and virtualisation layers.

Several derived metrics are used in specific stages, and three of them deserve an explicit caveat because they are *not* directly measured quantities even though they appear as numbers in result tables.

The *boundary-crossing interval* is a composite that bundles serialisation, transport, and the work executed on the remote side after the boundary. It is not the cost of "crossing the wire" alone – the thesis does not instrument the gRPC stack internally – but the time observed at the client between dispatching the request and receiving the response, minus the local pre-boundary compute. Any interpretation as a "transport cost" alone is therefore unsafe; it is a from-here-to-there interval, not a pure transport measurement.

The *activation-transfer burden* is derived from the model architecture as the intermediate tensor size at a split boundary (or the cumulative tensor size across all boundaries in a chain). It is an architectural quantity, not a measured one: no network profiler is used to confirm the on-wire byte count. The 16 MiB gRPC message limit and the parity validation against the monolithic model together ensure that the architectural size is what is actually transmitted, but the figure itself is computed from model shapes rather than from packet capture.

The *inferred non-compute / platform cost* is an analytical residual:

$$\widehat{\text{non-compute}} = \text{end-to-end} - \widehat{\text{compute}},$$

where the compute estimate comes from per-segment local timing. It is used in the Kubernetes, AKS, and mTLS analyses to separate platform and communication overhead from model computation. As a residual it inherits the error of the compute estimate; it cannot be falsified against a directly measured ground truth and should be interpreted as an upper bound on the platform + communication share rather than as a direct measurement of it.

In RQ2.1, resource and operational overhead metrics are also reported, including sidecar CPU and memory, total pod resource deltas, schedule-to-ready time, and the count of added mesh and security objects. In RQ2.2, sequential throughput is derived from mean latency as requests per second to express the throughput impact of confidential-node placement; this is an algebraic restatement of mean latency at batch size one, not an independent throughput measurement under load.

4.5.2 Cross-Round Consistency

Cross-round consistency is assessed using the round CV, computed as the standard deviation of per-round condition means divided by the grand mean. A low round CV indicates stable execution across independent rounds and supports treating the reported condition means as representative rather than as artefacts of a single favourable or

unfavourable run. Cross-round stability and parity validation are treated as the primary indicators of methodological validity.

4.5.3 Effect Size and Significance Tests

In stages with 1,000 per-condition measured iterations, Cohen's d is computed from pooled per-iteration data and the Mann-Whitney U test is reported for completeness. These statistics are descriptive diagnostics rather than the main decision rule. The thesis bases methodological carry-forward decisions on the predeclared eligibility and near-best-window rules in section 4.6; statistical-test outputs are used to contextualise the size and consistency of observed differences.

4.6 Carry-Forward and Selection Logic

The staged pipeline uses a carry-forward mechanism for the local split-selection stages (RQ1.1 and RQ1.2). After each of these stages, one or two candidates are selected to enter the next stage according to two predeclared rules.

The first rule concerns compute degeneracy. A split condition is classified as compute-degenerate if the smaller of the two services contributes less than 10% of the total split compute, i.e. $\min(t_a, t_b)/(t_a + t_b) < 0.10$, where t_a and t_b are the measured per-segment compute times. Compute-degenerate conditions are excluded from main-candidate eligibility because they produce structurally unbalanced decompositions that do not constitute meaningful microservice partitions, even if their raw latency is low.

The second rule selects two candidates from the eligible (non-degenerate) set inside the 5% near-best window: the condition with the lowest mean end-to-end latency is selected as the main carry-forward candidate, and the next-lowest is retained as the reference candidate. Ties on mean latency are broken in favour of the smaller activation-transfer size, so that activation size acts as a structural tie-breaker rather than as a primary selection criterion.

Carry-forward is not applicable in RQ1.4 and RQ1.5, which benchmark a predefined chain family rather than selecting among candidates. It is also not applicable in the RQ2 stages, which fix the deployment topology established by RQ1 and vary only the security configuration.

4.7 Stage-Specific Methods

4.7.1 RQ1.1 – Coarse Architectural Boundary Selection

RQ1.1 screens the four coarse stage boundaries of ResNet-18 against a monolithic baseline under the maximally controlled local setup described in Sections 4.4 and 4.3. The four split conditions place the boundary after `layer1`, `layer2`, `layer3`, and `layer4`, producing intermediate activation tensors of approximately 784 KiB, 392 KiB, 196 KiB, and 98 KiB in FP32 respectively. Both split services run as separate OS processes communicating via gRPC over `127.0.0.1:50051`.

The justification for screening at the coarse architectural level before conducting any finer search is that moving directly into an exhaustive layer-by-layer search would risk

over-fitting the selection to incidental local-execution noise. Prior DNN-partitioning systems commonly reason about split placement at layer or candidate-boundary granularity [15, 18], while later work motivates reducing the candidate set when exhaustive layer-wise search becomes impractical [18, 20, 33]. This thesis therefore uses stage boundaries for the first screening pass as a deliberate search-space reduction and only refines inside the carried-forward region in RQ1.2.

The output of this stage is one main carry-forward candidate and one reference candidate, selected using the degeneracy and near-best-window rules defined in Section 4.6.

4.7.2 RQ1.2 – Fine-Grained Refinement

RQ1.2 takes the carry-forward outcome of RQ1.1 as its starting point and introduces one block-level split point between the two coarse carry-forward anchors. The new condition, `split_after_layer3_block0`, bisects the `layer3` stage at its internal residual-block boundary.

A key structural property motivates this particular design choice. `split_after_layer3_block0` and `split_after_layer3` both produce intermediate activation tensors of approximately 196 KiB. The design therefore holds activation-transfer size constant while varying compute placement, creating a controlled comparison that can isolate the role of compute distribution independently of communication burden. This comparison is grounded in the DNN-partitioning literature, which treats compute distribution as a first-class criterion alongside activation-transfer cost [18, 28].

The experimental setup and measurement protocol are identical to RQ1.1, ensuring that the two stages remain directly comparable.

4.7.3 RQ1.3 – Explanatory Synthesis

RQ1.3 is not a split benchmark. It is a hybrid analytical stage that explains the patterns observed in RQ1.1 and RQ1.2 using two structural properties: activation-transfer size and compute distribution. The explanatory analysis draws on three evidence sources: the split benchmark summaries from RQ1.1, the split benchmark summaries from RQ1.2, and a separately conducted internal timing experiment on the monolithic ResNet-18 model.

The internal timing experiment instruments the monolithic forward pass at the stage, block, and selected operation level. It uses the same single-threaded CPU-stabilised environment as the split benchmarks, with 10 rounds, 50 warmup iterations per round, and 200 measured iterations per round. Functional equivalence against the uninstrumented model is verified ($\texttt{max\_abs\_diff} = 0.0$), and instrumentation overhead is measured explicitly at approximately 0.71 ms (0.94% of total forward-pass time), confirming that profiling does not materially distort the timing distribution.

The justification for this supporting experiment is that split-benchmark boundary-crossing intervals reflect a composite of serialisation, transport, and remote-side computation. Disentangling those components – particularly distinguishing activation-transfer cost from compute-placement effects – requires a direct measurement of local per-block execution cost that the split benchmark alone cannot provide.

4.7.4 RQ1.4 – Multi-Microservice Kubernetes Chain

RQ1.4 moves from local process-based communication to real Kubernetes in-cluster services. The environment is a single-node `kind` cluster running on Ubuntu/Linux, with all service pods colocated on the same node. This colocation rule is applied deliberately so that inter-service gRPC traffic remains in-cluster rather than traversing an external network, meaning that any observed overhead reflects orchestration and pod-to-pod communication rather than wide-area latency.

The stage benchmarks a predefined family of five conditions from `monolithic_k8s_1svc` to `chain_5svc`, corresponding to one through five synchronously chained ResNet-18 service segments. The chain family is predefined rather than selected by carry-forward because the purpose of this stage is not to choose among candidates but to characterise the full cost curve as a function of service count. The split points used are: `layer2` for chain_2svc; `layer1`, `layer3` for chain_3svc; `layer1`, `layer2`, `layer3` for chain_4svc; and `layer1`, `layer2`, `layer3`, `layer4` for chain_5svc. This family deliberately spans the full range of RQ1.1 activation sizes from 784 KiB (`layer1`) down to 98 KiB (`layer4`), and consequently re-introduces boundaries that RQ1.1 rejected (`layer1` as too transfer-heavy and `layer4` as compute-degenerate). The resulting cost curve should therefore be read as an *upper-bound envelope* on the cost of adding service boundaries – a purely best-boundary chain at each depth would lie below this envelope – not as a generic per-hop cost. Service pods are allocated one CPU core and 512 MiB of memory with Guaranteed QoS to fix the resource envelope and reduce scheduler and eviction variability during measurement windows.

The use of a single-node `kind` cluster is an acknowledged limitation: it eliminates inter-node network hops and does not capture the scheduling and networking variability of a multi-node or managed cluster. This limitation is addressed in three stages: RQ1.4b adds a `kind` CNI-bridge sensitivity probe (still single-host, so it does not measure real inter-node networking), RQ1.5 re-runs the same chain family on managed AKS to validate ordering, and RQ1.5b adds a genuine multi-node sensitivity stage on a six-VM AKS cluster so the real inter-node penalty becomes a measured, bounded quantity.

4.7.5 RQ1.4b – kind CNI Sensitivity (appendix-only)

RQ1.4b is a supporting sensitivity stage that is reported as appendix-level evidence only, not as a headline number.

The stage re-runs the same five-condition chain family from RQ1.4 on a six-worker `kind` cluster with pod anti-affinity enforced so that every chain segment of a given condition lands on a distinct kind worker node and the benchmark client lands on a sixth dedicated worker. The shared benchmark contract is otherwise unchanged (same model, same Guaranteed-QoS pod profile, same five-round / 200-measured-iteration / 50-warmup discipline, same image tag).

A methodological caveat governs how this stage should be interpreted. kind's "nodes" are Docker containers that share a kernel on a single host machine and communicate over kind's CNI bridge; they are not separate physical or virtual hosts. RQ1.4b therefore measures the cost of routing inter-service gRPC through the kind CNI bridge under anti-affinity, rather than the cost of real cross-host networking. The genuine multi-

host inter-node penalty is measured separately in RQ1.5b on a multi-node AKS cluster (section 4.7.7). Round-level cross-condition consistency (round CV) is also materially higher in RQ1.4b than in single-node RQ1.4, reflecting kind-CNI variability rather than a property of the chain family. Comparisons against RQ1.4 use the per-condition penalty $\Delta_c = \text{mean}_c^{\text{kind-cni}} - \text{mean}_c^{\text{single}}$, which is read as a kind-CNI sensitivity bound rather than as a deployment-grade multi-node result.

4.7.6 RQ1.5 – AKS Transfer Validation

RQ1.5 re-runs the same predefined chain family from RQ1.4 on a single-node AKS cluster (`Standard_D8s_v3`, region `swedencentral`, `kubenet` networking). The stage does not aim to replicate the local Kubernetes absolute latency values, which are expected to differ because the two environments use different host CPUs, kernels, virtualisation layers, container runtimes, and networking paths.

The transfer claim is therefore directional rather than numerical: if the same architectural chain ordering is preserved in AKS, that constitutes evidence that the ordering reflects the architecture of the model partition rather than incidental local-cluster noise. This emphasis on within-environment comparison is consistent with benchmarking guidance that absolute performance numbers do not transfer across hardware and software stacks without careful environment characterisation [13]. Within-environment normalised overhead is used as the primary comparison metric for this reason.

All service pods and the benchmark client are colocated on the same AKS node and run with Guaranteed QoS, applying the same single-node placement discipline as RQ1.4. Pod resource requests differ between the two stages: RQ1.4 service pods request 1 CPU and 512 MiB of memory, whereas RQ1.5 service pods request 1 CPU and 1 GiB, matching the larger memory headroom available on `Standard_D8s_v3`. This difference is documented per artefact and does not affect placement or QoS class.

4.7.7 RQ1.5b – AKS Multi-Node Sensitivity

RQ1.5b is a sensitivity stage that quantifies the inter-node network cost the single-node RQ1.5 design intentionally hides. The stage re-runs the same predefined chain family on a six-node AKS cluster (`Standard_D8s_v3` $\times$ 6, region `swedencentral`, single node pool, single VNet) and enforces *multi-node* placement: each chain segment of a given condition lands on a distinct cluster node, and the benchmark client lands on a sixth dedicated node. Placement is enforced by Kubernetes pod anti-affinity keyed on `kubernetes.io/hostname`, with two terms per service deployment: a service-vs-service term using the `experiment-condition` and `workload-role: service` labels, and a service-vs-client term using the `workload-role: client` label. The benchmark client carries a mirror anti-affinity rule against `workload-role: service`.

Compliance is verified after deployment by reading observed `nodeName` values from the Kubernetes API. A condition is admitted to the rolling merge only if its service pods are pairwise disjoint by node and the client node is distinct from every service node; a single collision aborts the run. The shared benchmark contract is otherwise unchanged from RQ1.5: same model, same chain family, same Guaranteed-QoS pod profile, same five-round / 200-iteration / 50-warmup discipline, and the same image used in the same-day

RQ1.5 single-node run. The headline sensitivity metric is the per-condition multi-node penalty $\Delta_c = \text{mean}_c^{\text{multi}} - \text{mean}_c^{\text{single}}$, computed against the same-day frozen RQ1.5 single-node baseline so that the inter-node delta is not contaminated by region-time drift.

4.7.8 RQ2.1 – Service Identity and Mutual TLS Hardening

RQ2.1 introduces a paired execution design. Each paired pass executes both the plain AKS condition and the mTLS/AuthZ condition in alternating order, with the order determined by a seeded plan in which three passes run mTLS first and two passes run plain first. This alternation reduces the risk that the observed delta is explained by slow performance drift over the course of the run rather than by the hardening mechanism itself.

The plain condition uses standard Kubernetes service-to-service communication without mesh injection. The mTLS condition adds one Istio sidecar proxy per inference service, service accounts, workload-level peer-authentication objects, and authorisation-policy objects. These mechanisms correspond to the common Istio pattern of using sidecar proxies for service-to-service mTLS and policy enforcement in Kubernetes microservices [4]. The benchmark client remains outside the mesh; the stage therefore measures hardening of the inter-service path inside the inference chain, not external ingress hardening.

The managed AKS Istio add-on (`asm-1-29`) is used as the mesh implementation. This choice reflects the thesis's focus on realistic cloud-native deployment: managed add-ons represent the configuration path most likely to be used by practitioners adopting mTLS in AKS and their behaviour reflects actual cloud-operator defaults rather than custom tuning.

Security validation is performed for each paired run using four methods: static manifest validation of peer-authentication and authorisation-policy objects; runtime validation of policy object counts; a full-chain positive probe confirming that valid upstream-to- downstream requests succeed; and a direct non-mesh denial probe confirming that unauthenticated access to protected downstream services is blocked.

The primary topology is `chain_2svc` (split after `layer2`). It is the lowest-overhead non-monolithic chain in the preceding RQ1.4 / RQ1.5 stages. The split point used is `layer2`, which corresponds to the RQ1.1 *reference* carry-forward candidate rather than the main carry-forward (`layer3`); this is a deliberate choice because `chain_2svc` in the RQ1.4 / RQ1.5 chain family is built on `layer2`, so reusing the same boundary preserves continuity with the already-frozen chain-cost numbers.

A `chain_5svc` stress topology is included to determine how mTLS overhead grows when more service boundaries are protected. `chain_5svc` reuses the same RQ1.4 / RQ1.5 split family (`layer1`, `layer2`, `layer3`, `layer4`), which deliberately includes the RQ1.1-rejected boundaries to span the full activation range. The chain_5svc overhead figure should therefore be read against the same upper-bound-envelope caveat introduced in section 4.7.4.

A known limitation of sidecar-based mTLS is that the measured protection boundary is the proxy-mediated inter-service data path, not all data handling inside the pod or application process. MeshInsight shows that sidecar meshes add app-to-sidecar and sidecar-to-sidecar data-path segments whose latency and resource costs depend on

proxy configuration and workload details [36]. A second limitation is that the mesh control plane and certificate authority remain trust-critical components even when the data-plane is fully mTLS-protected [27]. Both limitations are acknowledged explicitly: RQ2.1 is claimed as a communication-level hardening layer, not as a complete runtime-secrecy mechanism.

4.7.9 RQ2.1 (Ablation) – Decomposition of the Hardening Bundle

RQ2.1's main paired design treats the mTLS configuration as a single bundle: sidecar injection, strict peer-authentication, and authorisation-policy enforcement are all applied together. An additional ablation campaign is run on `chain_2svc` to decompose this bundle into its incremental contributions.

The ablation introduces four conditions on the same AKS cluster, same Istio revision (`asm-1-29`), and same strict same-node placement as the primary RQ2.1 design: (c0) plain non-mesh; (c1) mesh-default sidecar with automatic mTLS but no AuthorizationPolicy; (c2) strict PeerAuthentication mTLS with no AuthorizationPolicy; and (c3) strict mTLS with AuthorizationPolicy. The conditions are run as a seeded interleaved campaign with five passes per condition (5 passes $\times$ 4 conditions $\times$ 200 measured iterations = 4,000 measured rows).

The adjacent deltas decompose the hardened path as follows. $c3 - c0$ is the *full hardened-path cost* and $c3 - c2$ is the *clean AuthorizationPolicy cost*; both are cleanly attributable. $c1 - c0$ and $c2 - c1$ require a methodological caveat: Istio's mesh-default behaviour is automatic mTLS in PERMISSIVE mode, so c1 already encrypts sidecar-to-sidecar traffic. The delta $c1 - c0$ therefore measures the combined cost of *sidecar process + IPC + auto-mTLS data plane* over plain non-mesh, and cannot be attributed to "sidecar process overhead" alone. The delta $c2 - c1$ measures the cost of upgrading the policy from PERMISSIVE to STRICT with auto-mTLS already in effect, not the cost of "strict mTLS encryption" relative to plaintext sidecar traffic. A clean separation of sidecar process cost from mTLS cryptography cost would require a fifth condition with the sidecar present but `PeerAuthentication: DISABLE`; this thesis does not include that condition and treats the disentanglement as future work.

This campaign is treated as supporting, not headline. The ablation deltas are internally comparable across c0–c3 within the same campaign run, but the absolute numbers must not be spliced into the frozen two-condition paired `chain_2svc` and `chain_5svc` results.

4.7.10 RQ2.1 (Split Sensitivity) – Single-Hop mTLS vs Activation Size

A second supporting campaign measures whether the mTLS overhead at a single protected hop scales with the size of the activation crossing that hop. The independent variable in this campaign is the split point that produces hop 2's activation, varying it across `layer1`, `layer2`, `layer3`, and `layer4` (activation sizes 802 816, 401 408, 200 704, and 100 352 bytes respectively, matching the RQ1.1 sizes).

For each split point, a plain condition and an mTLS condition are run as a paired pair, yielding eight conditions in total. Only the hop between service 1 and service 2 is

protected by the mTLS configuration; the same strict same-node placement, Istio revision, pod profile, and image digest as the primary RQ2.1 design are otherwise preserved. The campaign runs five seeded-interleaved passes across the eight conditions for a total of 8,000 measured rows. The per-split deltas (mean and p95 mTLS overhead, per-split operational overhead, and per-split resource summary) are the headline outputs.

This campaign is treated as supporting, not headline, for two reasons. First, the `split_after_layer4` condition remains compute-degenerate per the RQ1.1 degeneracy rule and is retained only as a low-payload diagnostic endpoint, not as a candidate deployment. Second, the per-split mTLS deltas are campaign-internal: they may not be spliced into the frozen `chain_2svc` or `chain_5svc` paired numbers.

4.7.11 RQ2.1b – Multi-Node Sensitivity of mTLS Overhead

RQ2.1b is a paired sensitivity stage that quantifies how the chain_2svc mTLS overhead changes when the two inference services are forced onto distinct AKS nodes. It mirrors the RQ1.5b design at the security layer: the `chain_2svc` plain and mTLS conditions are deployed on a dedicated AKS node pool with `multi_node_anti_affinity` placement and a dedicated benchmark-client node. The same paired alternation, five-pass budget (5 paired passes $\times$ 2 conditions $\times$ 200 measured iterations $= 2{,}000$ rows), seeded order, and security-validation contract as the primary RQ2.1 chain2 run are reused.

Compliance with the multi-node placement contract is verified after deployment by reading observed `nodeName` values from the Kubernetes API and rejecting any pass where the two service pods share a node or where the client node coincides with either service node. The headline output is the multi-node paired mTLS delta (mean, p95, and per-pass spread) together with the placement-validation status of every pass.

4.7.12 RQ2.2 – Confidential VM Execution Hardening

RQ2.2 keeps the communication-security contract from RQ2.1 fixed and changes the execution environment of the downstream protected service. The confidential condition places `service2` on an AMD SEV-SNP confidential node pool (`Standard_DC8as_v5`); the standard condition uses matched non-confidential AMD nodes (`Standard_D8as_v5`) in the same AKS cluster.

The RQ2.2 cluster is deployed in Azure region `westeurope` rather than the `swedencentral` region used by the RQ1.5, RQ1.5b, and RQ2.1 clusters, because at the time of the experiments the AMD SEV-SNP DC-series SKUs (`Standard_DC8as_v5`) were the constraining resource and were available with the required quota in `westeurope`. This region change is the reason a fresh paired standard baseline is collected on the RQ2.2 cluster rather than reusing the RQ1.5 or RQ2.1 mTLS baseline: absolute latency between the two regions is not directly comparable, so only the within-cluster paired delta between the standard and confidential conditions is reported as the incremental TEE overhead. This paired design reduces region- and session-level confounding, although the delta still reflects the practical platform change from the standard AMD node pool to the confidential AMD SEV-SNP node pool.

The thesis-facing TEE scope is `service2_only`. This selective placement isolates the cost of protecting the downstream service that receives intermediate activations,

motivated by activation-privacy work [12] showing that those activations are recoverable when held by an untrusted remote partition. Extending the confidential-execution boundary to both services (full two-service TEE) is out of the thesis-facing scope of RQ2.2 per the evidence manifest and is treated as future work rather than measured here.

The protection boundary of AMD SEV-SNP must be stated precisely. SEV-SNP provides VM-level memory encryption and integrity protection against an untrusted host platform [3, 17], but the guest operating system, application process, PyTorch runtime, model weights, and Istio sidecar all remain inside the same VM trust boundary. Recent work additionally shows residual attack surfaces beyond memory encryption, including malicious-interrupt injection from the untrusted hypervisor [29]; this thesis does not experimentally evaluate such attacks. The motivation for protecting the downstream segment at infrastructure level is supplied by activation-privacy work [12] showing that intermediate feature maps can be inverted to recover client inputs, although that threat model differs from the host-level adversary SEV-SNP defends against. RQ2.2 therefore uses the term *VM-level confidential execution* throughout rather than claiming application-level enclave isolation.

4.7.13 RQ2.3 – Security-Hardening Trade-off Synthesis

RQ2.3 is a synthesis stage: no new benchmark data is collected. The stage combines the plain AKS baseline from RQ1.5, the mTLS/AuthZ results from RQ2.1, and the selective `service2_only` TEE result from RQ2.2 to compare the three thesis-facing hardening configurations across measured cost, operational complexity, and protection scope. Full two-service TEE is treated as future work rather than as a measured trade-off point.

The synthesis is structured around the principle that the preferred hardening configuration is threat-model-dependent rather than universally optimal. Applying every available security mechanism to every service boundary is not always the best engineering choice, because each additional protection layer adds measurable latency, resource, and operational cost. RQ2.3 therefore frames its output as a structured trade-off comparison rather than a single global ranking, allowing a practitioner to match a security configuration to a stated threat model and performance budget.

4.8 Artifact Freezing and Reproducibility

Thesis-facing runs are exported and frozen as artifact sets before analysis is performed. This means that the reported results are derived from a fixed, immutable snapshot of the raw measurement data rather than from ad hoc terminal output or re-runnable live scripts. Each frozen artifact set records the full execution environment: operating system, Python and PyTorch version, Kubernetes version and cluster name, node type, namespace, container image reference, condition ordering, and parity validation outcomes. For the AKS stages (RQ1.5, RQ1.5b, RQ2.1, RQ2.1b, RQ2.2) the image reference is a pinned `sha256` digest from the project's Azure Container Registry. For the local `kind` stages (RQ1.4, RQ1.4b) the image reference is a mutable tag (`thesis-inference:latest`) preloaded into the cluster with `IfNotPresent`; reproduction in those stages therefore depends on the locally built image alongside the recorded runtime metadata.

The separation between the frozen artifact and the analysis code ensures that the reported numbers remain stable even if benchmark scripts are modified for subsequent exploratory runs. The repository contains both thesis-facing configs and older exploratory or smoke-test configs; the artifact freeze identifies which run is the thesis-facing primary result.

4.9 Limitations and Threats to Validity

The methodological scope of this study has known limitations covering five themes, each of which constrains how the results may be generalised.

Single-node deployment. The primary RQ1.4 and RQ1.5 designs colocate all chain services on one node so that the measured cost is attributable to orchestration and gRPC rather than to wide-area or inter-node networking. RQ1.4b and RQ1.5b are explicit multi-node sensitivity stages that bound the cost the single-node design hides; the RQ1.5/RQ2.1 primary numbers should not be read as representative of arbitrary multi-node deployments.

CPU-only, single-model, single-workload scope. All experiments use ResNet-18 in FP32 on CPU at batch size one. Results may not transfer quantitatively to GPU inference, mixed precision, larger or smaller models, transformer architectures, or batched throughput-oriented workloads, although the staged methodology itself is model-agnostic.

Controlled-versus-production variability. Strict CPU stabilisation is enforceable on the local stages but only partially enforceable on AKS (governor and turbo are not exposed; `nice` is denied). The measurement environment is therefore quieter than a typical production AKS workload, and absolute numbers may shift under contended noisy-neighbour conditions.

Bounded RQ2.1 threat model. RQ2.1 measures communication-level hardening only. The sidecar design protects the proxy-mediated inter-service path, while data handling inside each pod and application process remains outside the measured mesh boundary; the mesh control plane plus CA also remain trust-critical. RQ2.1 must therefore be interpreted as a confidentiality and authenticity boundary on the inter-service path, not as a complete runtime-secrecy mechanism.

Bounded RQ2.2 threat model. RQ2.2 measures the incremental cost of AMD SEV-SNP at VM granularity. The guest OS, application, runtime, model weights, and sidecar remain inside the VM trust boundary, and residual attack surfaces such as malicious-interrupt injection from the untrusted hypervisor are out of experimental scope. RQ2.2 is therefore an infrastructure-level confidentiality control against a host-platform adversary, not an application-level enclave guarantee.

In addition to the five themes above, several measurement scopes are deliberately out of this thesis and should be named explicitly so that the reported numbers are not over-generalised. *Concurrency.* All measurements use batch size one and sequential request issuance; throughput under concurrent load is not measured, and the cost of mTLS, sidecar IPC, and SEV-SNP memory encryption may scale non-linearly when the sidecar or the CVM memory controller is contended. *Tail behaviour beyond p95.* With 1,000 measured iterations per condition the p95 estimate is well populated, but only ten samples sit beyond the p99 cut and the data do not support reliable claims about p99 or deeper tail percent-

iles. *Cold start and steady state.* Pod startup, sidecar bootstrap, CVM boot, and SEV-SNP attestation latency are not part of the measured end-to-end interval; the headline numbers describe steady-state behaviour only. *Cloud-dollar cost.* The thesis reports resource overhead in CPU-time and memory units rather than in Azure billing units; converting to dollar-equivalents at the SKU prices used (`Standard_D8s_v3`, `Standard_D8as_v5`, `Standard_DC8as_v5`) is left to the synthesis discussion in RQ2.3. *Sidecar capacity headroom.* The Istio sidecar limits are taken from the managed-mesh defaults; whether the sidecar approaches its CPU limit at the measured load is checked via the resource sampling captured in the RQ2.1 artefacts and is reported alongside the overhead numbers, but no load-stress condition was run to find the sidecar's saturation point.

The consolidated discussion of these limitations – including their implications for the synthesis in RQ2.3 and for external validity – is in section 6.6.

4.10 Ethical Considerations

This thesis does not involve human participants, personal data, or biological material. The study is a software systems benchmark using a publicly available pretrained image classification model and standard synthetic inputs; no classification decisions derived from this model are used to affect any real-world outcomes, and no personal images or sensitive datasets are processed.

ResNet-18 with pretrained ImageNet weights inherits any biases or limitations of the ImageNet training process. However, since the thesis does not evaluate model accuracy and does not deploy the model in a decision-making context, these concerns are not directly relevant to the research questions addressed here.

Cloud compute resources are used responsibly: experiments are scoped to the minimum infrastructure required to answer each research question, and resources are deallocated after each benchmark stage. All measurement artifacts contain only timing data and system metadata; no sensitive data is stored on cloud infrastructure.

Chapter 5

Experiments & Results

5.1 RQ1.1 Coarse Architectural Boundary Selection

5.1.1 Research Question

This subsection addresses the following sub-research question:

> *Which coarse architectural stage boundaries provide the most suitable two-part microservice decompositions of ResNet-18 under controlled local execution, and how do their latency and boundary-crossing costs compare to monolithic inference?*

RQ1.1 is designed as a coarse-boundary screening stage rather than an exhaustive exploration of all possible split points. Its purpose is to identify architecturally meaningful two-part decompositions that can be compared against monolithic inference under controlled local execution, and to determine which coarse boundaries provide the strongest carry-forward basis for later refinement. In this stage, suitability is not treated as a question of raw latency alone. It is evaluated in relation to the joint trade-off between end-to-end latency, activation-transfer burden, boundary-crossing cost, and the structural balance of computation across the two resulting services.

The literature framing for coarse architectural boundary selection is discussed in sections 3.1 and 3.2.

5.1.2 Experimental Setup

The RQ1.1 benchmark used the fully controlled local CPU-only configuration in order to minimize variance and isolate the effect of the partition boundary itself. The source configuration file is `configs/rq1/1.1/rq1_1_fully_controlled.yaml`; an exact byte-for-byte snapshot is preserved alongside the frozen results as `config.yaml` in the artifact directory. The evaluated model was ResNet-18 with pretrained weights, executed on CPU in FP32 precision with an input shape of $1 \times 3 \times 224 \times 224$. The benchmark compared one monolithic baseline against four two-part split configurations, each corresponding to a coarse architectural stage boundary in ResNet-18.

Activation-tensor sizes in this section are quoted in binary kibibytes (KiB, 1024 bytes) for consistency with the gRPC message-size limit, which is also a binary quantity. Three further configuration files are present under `configs/rq1/1.1/` but do not contribute to the local-controlled RQ1.1 thesis-facing result reported here: `rq1_1_experimental.yaml` (a smoke test with minimal iterations, used only for pipeline validation), `rq1_1_threaded.yaml` (a four-core ancillary profile retained as a historical comparison point), and `rq1_1_azure_fully_controlled.yam` (an AKS-backed coarse-split variant driven by `scripts/run_rq11_azure_fully_controlled.py`, retained for completeness but not part of the local single-host result reported in this section).

The benchmark was configured to use a single-threaded PyTorch execution model pinned to one physical CPU core with SMT avoided. Thread counts for PyTorch, OpenMP, MKL, and OpenBLAS were all fixed to one. In addition, process priority was increased, the CPU governor was set to `performance`, and turbo boost was disabled in order to reduce frequency-related variance. These controls were applied symmetrically across all conditions so that the comparison remained fair and reproducible.

Table 5.1: RQ1.1 benchmark configuration

Setting	Value
Model	ResNet-18 (pretrained)
Device	CPU
Precision	FP32
Input shape	$1 \times 3 \times 224 \times 224$
Conditions	Monolithic + split after layer1, layer2, layer3, layer4
Rounds	5
Warmup iterations	50
Measured iterations	200
Cooldown between conditions	5 seconds
Random seed	42
Communication	gRPC over localhost (`127.0.0.1:50051`)
Maximum message size	16 MiB (16777216 bytes)
Parity tolerance	1.0×10^{-5}
Parity inputs	5
Warmup window	10 iterations
Warmup CV threshold	0.02

5.1.3 Benchmark Design

The experiment followed a repeated-measures design with five conditions: a monolithic baseline and four split configurations. The split points were selected after the major architectural stages of ResNet-18, namely after `layer1`, `layer2`, `layer3`, and `layer4`. These boundaries are meaningful because they preserve the stage structure of the network while allowing the evaluation of progressively later partition points.

The benchmark protocol consisted of 5 rounds. For each round and condition, 50 warmup iterations were executed before 200 measured iterations were recorded. A 5-second cooldown separated successive conditions. The ordering of conditions was randomized using deterministic per-round seeds derived from the configured seed and round index to reduce ordering bias. Functional equivalence between monolithic and split executions was enforced as a mandatory precondition through both local and gRPC parity checks using five inputs and an absolute tolerance of 1.0×10^{-5}.

A warmup calibration procedure was used to monitor whether latency had stabilized. This procedure used a trailing window of 10 iterations and a coefficient-of-variation threshold of 0.02. The implementation could extend warmup adaptively beyond the configured minimum if the trailing window remained unstable; in this frozen run, all condition-round pairs stabilized within the configured 50 iterations, so no extra warmup iterations were used.

5.1.4 Partition Conditions

The four split conditions corresponded to the following coarse stage boundaries in ResNet-18:

- **split_after_layer1:** The first service executes the stem and `layer1`, and the second service executes `layer2` through the classifier head. This split transfers the largest intermediate tensor, of exact shape $1 \times 64 \times 56 \times 56$, corresponding to $802{,}816$ bytes (784 KiB) in FP32.
- **split_after_layer2:** The first service executes the stem and `layer1--layer2`, and the second service executes `layer3` through the classifier head. The intermediate tensor is of exact shape $1 \times 128 \times 28 \times 28$, $401{,}408$ bytes (392 KiB).
- **split_after_layer3:** The first service executes the stem and `layer1--layer3`, and the second service executes `layer4` plus pooling and classification. The intermediate tensor is of exact shape $1 \times 256 \times 14 \times 14$, $200{,}704$ bytes (196 KiB).
- **split_after_layer4:** The first service executes the stem and `layer1--layer4`, and the second service executes only average pooling and the final linear layer. The intermediate tensor is of exact shape $1 \times 512 \times 7 \times 7$, $100{,}352$ bytes (98 KiB).

These stage boundaries provide a coarse sweep across the network from earlier to later partition points. This design is consistent with partitioning research that treats split placement as a search-space and granularity decision shaped by computation, communication, and deployment constraints, rather than as an unconstrained exhaustive sweep over every internal operation [18, 20, 33].

Carry-forward rule. The decision of which coarse boundary to promote to the next stage is governed by a predeclared two-part rule applied to the per-condition mean end-to-end latencies and the per-service compute measurements (see section 4.6 for the full statement). In short: (i) the *near-best window* comprises every split whose mean latency lies within 5% of the raw-fastest split; (ii) a split is treated as *compute-degenerate* when the smaller of the two per-service mean compute times contributes less than 10% of the total split compute, i.e. $\min(A, B)/(A + B) < 0.10$ for service-A compute A and service-B compute B. The main carry-forward candidate is the non-degenerate split with the lowest mean latency in the near-best window; the next-best non-degenerate split is retained as the reference candidate. The frozen rule outputs for this run are recorded in `carry_forward.json`.

Boundary-crossing interval. The *boundary-crossing interval* reported for each split condition is the client-observed interval that begins immediately before the intermediate-tensor serialisation and ends immediately after the response deserialisation. It is a composite of activation serialisation, gRPC transport, the remote service-B compute, and response deserialisation, and is implemented by the timer block in `src/client/split_client.py` (see section 4.3 for the full measurement contract). It is not a pure "wire" cost.

5.1.5 Results

The thesis-facing RQ1.1 benchmark used the frozen artifact `rq1_1_20260515_111428` (source: `results/frozen_new/rq1_1_20260515_111428/condition_summaries.csv`). Monolithic execution achieved the lowest mean end-to-end latency at 77.24 ms. The four split conditions yielded mean latencies of 80.88 ms for `split_after_layer1`, 80.12 ms for `split_after_layer2`, 79.08 ms for `split_after_layer3`, and 79.58 ms

for `split_after_layer4`. This corresponds to relative overheads of approximately 4.7%, 3.7%, 2.4%, and 3.0%, respectively, compared to monolithic execution.

Table 5.2: RQ1.1 summary of mean latency, dispersion, descriptive within-run per-iteration 95% confidence interval on the mean, and overhead vs. monolithic. Source: `condition_summaries.csv` in `rq1_1_20260515_111428`. Confidence intervals are reported from the same artifact's `ci95_lower_ms` / `ci95_upper_ms` columns and summarise per-iteration variation within this run.

Condition	Mean (ms)	Median (ms)	Std. Dev. (ms)	Within-run 95% CI (ms)	Overhead vs. Monolithic
Monolithic	77.24	75.29	3.59	$[77.01, 77.46]$	–
Split after layer1	80.88	80.24	2.27	$[80.74, 81.02]$	4.7%
Split after layer2	80.12	79.60	2.40	$[79.97, 80.27]$	3.7%
Split after layer3	79.08	78.38	2.31	$[78.94, 79.23]$	2.4%
Split after layer4	79.58	78.85	2.43	$[79.43, 79.73]$	3.0%

Figure 5.1 visualises the latency pattern reported in table 5.2. In this frozen run the raw-fastest split is the mid-to-late boundary after `layer3`, not the latest boundary, and the carry-forward decision still depends on both latency and structural compute balance rather than latency alone.

All split configurations therefore introduced positive latency overhead relative to monolithic inference. However, the magnitude and character of this overhead varied substantially with split location. The earliest split, after `layer1`, performed worst because it crossed the largest intermediate activation and exhibited the highest boundary-crossing cost. The latest split, after `layer4`, minimized activation-transfer size but left only a minimal residual computation segment on the second service (well under 1 ms of remote compute according to the per-service compute means in `condition_summaries.csv`), making it structurally compute-degenerate despite its low transfer burden. The mid-to-late splits, after `layer2` and `layer3`, formed the strongest non-degenerate candidates in the coarse sweep, with `split_after_layer3` also yielding the lowest raw split mean in this frozen run.

The boundary-crossing measurements further support this interpretation. The split after `layer1` incurred the largest boundary-crossing interval at 53.81 ms, followed by `layer2` at 40.85 ms and `layer3` at 26.06 ms, while `layer4` was much smaller at 2.22 ms. This descending pattern is consistent with the joint reduction in activation-transfer burden and downstream service-B compute as the boundary moves later, from 784 KiB at `layer1` to 98 KiB at `layer4`. At the same time, the late `layer4` split shows that minimizing transfer burden alone does not determine overall suitability as a microservice decomposition, because the downstream service becomes too small to represent a meaningful balanced distribution.

Cross-round consistency was strong across all configurations, indicating stable execution behavior. Per `round_consistency.json` for the same artifact, the monolithic condition had a round coefficient of variation (CV) of 0.002, while `split_after_layer1`, `split_after_layer2`, `split_after_layer3`, and `split_after_layer4` had round CVs of 0.005, 0.011, 0.003, and 0.005, respectively.

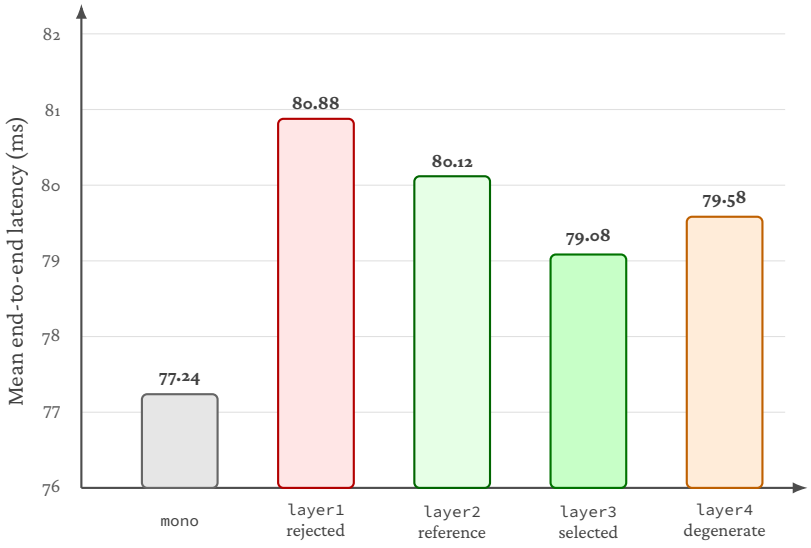


Figure 5.1: RQ1.1 coarse split latency results under controlled local execution. Bar heights are the mean end-to-end latency per condition for the frozen run `rq1_1_20260515_111428`; bar colours encode the role assigned by the carry-forward rule (baseline, selected_main, selected_reference, rejected, degenerate). `split_after_layer3` is the raw-fastest split in this run and is also non-degenerate; `split_after_layer4` is excluded by the compute-degeneracy rule despite its small intermediate activation. The carry-forward decision (see `carry_forward.json` for the same run) selects `split_after_layer3` as the main candidate and `split_after_layer2` as the reference candidate. The figure is generated deterministically from the frozen `condition_summaries.csv` and `carry_forward.json` by `scripts/render_rq11_figure.py`.

Scope caveat. The numbers reported in this section are derived from a single frozen artifact (`rq1_1_20260515_111428`) generated by a single benchmark run with the fixed seed 42. The five rounds within that run randomise condition order but do not vary the input seed, the host, or the build, and cross-host or cross-seed replication is not in scope for RQ1.1. The mean-latency differences between the non-degenerate split conditions are small relative to their per-iteration standard deviation. Although the descriptive within-run per-iteration 95% confidence intervals in table 5.2 do not overlap between `split_after_layer3` and `split_after_layer2`, those intervals should not be read as host-independent uncertainty estimates. The absolute ranking within the non-degenerate set should therefore be read as "the strongest carry-forward candidate observed in this frozen run" rather than as a general ordering. The carry-forward decision is stated in terms of the predeclared rule (section 4.6) applied to this artifact, and the ranking is carried forward for refinement in RQ1.2 rather than treated as a final answer.

The interpretation of these results and the answer to RQ1.1 are presented in section 6.1.1.

5.2 RQ1.2 Fine-Grained Refinement

5.2.1 Research Question

This subsection addresses the following sub-research question:

> *How does finer-grained refinement within the late-stage coarse region carried forward from RQ1.1 affect latency and communication-related overhead for two-part microservice inference?*

RQ1.2 is designed as a targeted refinement stage, not as a new blind search across the network. It takes the carry-forward outcome of RQ1.1 as its starting point. In RQ1.1, the coarse architectural boundary screening identified the mid-to-late region around `layer2` and `layer3` as the strongest balanced carry-forward zone, with `split_after_layer3` selected as the main candidate and `split_after_layer2` retained as the reference candidate. RQ1.2 therefore operates within that carried-forward comparison space and introduces one meaningful finer-grained split point between the two coarse anchors. The purpose is to determine whether a block-level boundary within this region reveals latency or communication-related differences that were not visible at the coarse stage level.

The literature framing for finer-grained partition refinement is discussed in section 3.2.

5.2.2 Experimental Setup

The RQ1.2 benchmark reused the same fully controlled local CPU-only configuration as RQ1.1 in order to ensure that the two stages remain directly comparable. The evaluated model was ResNet-18 with pretrained weights, executed on CPU in FP32 precision with an input shape of $1 \times 3 \times 224 \times 224$. Communication between the two split services used gRPC over localhost (`127.0.0.1:50051`) with a maximum message size of 16 MiB ($16{,}777{,}216$ bytes).

As in RQ1.1, activation-transfer sizes in this section are reported in binary kibibytes (KiB, 1024 bytes).

All CPU stabilisation controls from RQ1.1 were applied identically: single-threaded PyTorch execution pinned to one physical core with SMT avoided, thread counts for PyTorch, OpenMP, MKL, and OpenBLAS all fixed to one, process priority elevated, the CPU governor set to `performance`, and turbo boost disabled. These controls were applied symmetrically across all conditions so that environmental variation was reduced as a competing explanation for observed boundary-placement differences.

5.2.3 Benchmark Design

The experiment followed the same repeated-measures protocol as RQ1.1: 5 rounds of 200 measured iterations per condition, preceded by 50 warmup iterations and separated by 5-second cooldowns. Condition ordering was randomised within each round using a fixed seed. Functional equivalence between monolithic and split executions was enforced through both local and gRPC parity checks using five inputs and an absolute tolerance of

Table 5.3: RQ1.2 benchmark configuration

Setting	Value
Model	ResNet-18 (pretrained)
Device	CPU
Precision	FP32
Input shape	$1 \times 3 \times 224 \times 224$
Conditions	Monolithic + split after layer2, layer3.0, layer3
Rounds	5
Warmup iterations	50
Measured iterations	200
Cooldown between conditions	5 seconds
Random seed	42
Communication	gRPC over localhost (`127.0.0.1:50051`)
Maximum message size	16 MiB ($16{,}777{,}216$ bytes)
Parity tolerance	1.0×10^{-5}
Parity inputs	5
Warmup window	10 iterations
Warmup CV threshold	0.02

1.0×10^{-5}. A warmup calibration procedure using a trailing window of 10 iterations and a coefficient-of-variation threshold of 0.02 was applied to verify that latency had stabilised before measurement.

The key difference from RQ1.1 is the composition of the condition set. Rather than sweeping across all coarse stage boundaries, RQ1.2 evaluates a targeted set of three refined split conditions anchored by the RQ1.1 carry-forward candidates, with one new block-level split point inserted between them.

5.2.4 Refined Partition Conditions

The RQ1.2 condition set consists of one monolithic baseline and three split configurations:

- **split_after_layer2:** The first service executes the stem and `layer1`–`layer2`, and the second service executes `layer3` through the classifier head. This is the RQ1.1 reference carry-forward candidate. The intermediate activation tensor is approximately $128 \times 28 \times 28$, or 392 KiB in FP32.
- **split_after_layer3_blocko:** The first service executes the stem and `layer1`–`layer3.0` (i.e., the first residual block of `layer3`), and the second service executes from `layer3.1` onward through the classifier head. This is the single new finer-grained split point introduced in RQ1.2. It bisects the coarse `layer3` stage at its internal block boundary. The intermediate activation tensor is approximately $256 \times 14 \times 14$, or 196 KiB in FP32 — the same size as the tensor produced at the end of the full `layer3` stage.
- **split_after_layer3:** The first service executes the stem and `layer1`–`layer3`, and the second service executes `layer4` plus pooling and classification. This is the

Table 5.4: RQ1.2 summary of mean latency, overhead, and activation-transfer burden. Source files: `condition_summaries.csv`, `cross_condition.csv`, and `round_consistency.json` in the frozen artifact `rq1_2_20260515_112636`.

Condition	Mean (ms)	Median (ms)	Std. Dev. (ms)	Overhead	Activation (KiB)
Monolithic	77.565	75.641	3.663	—	—
Split after layer2	80.348	79.570	2.510	3.6%	392
Split after layer3_block0	80.157	79.379	2.564	3.3%	196
Split after layer3	80.392	79.687	2.475	3.6%	196

RQ1.1 main carry-forward candidate. The intermediate activation tensor is approximately $256 \times 14 \times 14$, or 196 KiB in FP32.

The monolithic baseline is retained for reference so that all overhead values remain anchored to the same absolute comparison. Together, these four conditions form a targeted refinement set rather than an exhaustive within-stage search: `split_after_layer2` and `split_after_layer3` anchor the carried-forward coarse space, while `split_after_layer3_block0` probes the single meaningful block-level internal boundary between them.

An important structural observation motivates this design. The coarse boundary after `layer3` and the block-level boundary after `layer3.0` produce intermediate activation tensors of the same size (both approximately 196 KiB). This creates a controlled comparison in which the activation-transfer burden is held constant while the distribution of computation across the two services differs. If the two conditions yield different end-to-end latencies or boundary-crossing intervals, the comparison points to compute placement as a plausible explanatory factor rather than to activation size alone; it does not by itself establish compute placement as the only source of the difference.

5.2.5 Results

The thesis-facing RQ1.2 benchmark used the tracked frozen artifact `rq1_2_20260515_112636`. Mean and table values use `condition_summaries.csv`; boundary and cross-round values use `cross_condition.csv` and `round_consistency.json` in the same artifact. The measured results are summarised in table 5.4. Monolithic execution achieved the lowest mean end-to-end latency at 77.565 ms. Among the three refined split conditions, `split_after_layer3_block0` achieved the lowest mean split latency at 80.157 ms, followed by `split_after_layer2` at 80.348 ms and `split_after_layer3` at 80.392 ms. This corresponds to overhead relative to monolithic execution of approximately 3.3%, 3.6%, and 3.6%, respectively. The three split means are tightly clustered within a 0.235 ms band in this run, so this ordering should be read as a narrow within-run ordering rather than as a materially separated ranking.

The boundary-crossing interval measurements reveal a notable difference between the two conditions that share the same activation size. `split_after_layer3_block0` exhibited a boundary-crossing interval of 33.97 ms, while `split_after_layer3` exhibited a boundary-crossing interval of 26.16 ms. Despite producing activation tensors of the same size (approximately 196 KiB), the two conditions differ by approximately 7.8 ms

in boundary-crossing cost. The earlier split (`layer2`) had the largest boundary-crossing interval at 40.86 ms, consistent with its larger activation tensor of 392 KiB.

Within-run cross-round consistency was strong across all conditions, per `round_consistency.json` in the same artifact. The monolithic condition had a round coefficient of variation (CV) of 0.0017, while the split conditions had round CVs of 0.0046 (`split_after_layer2`), 0.0033 (`split_after_layer3`), and 0.0070 (`split_after_layer3_block0`). The reported means can therefore be treated as representative of this frozen run rather than as artefacts of a single favourable round. This remains a single-run local result with fixed seed, host, and build; cross-seed or cross-host replication is outside the scope of RQ1.2.

The interpretation of these results and the answer to RQ1.2 are presented in section 6.1.2.

5.3 RQ1.3 Explanatory Synthesis: Activation-Transfer Size and Compute Distribution

5.3.1 Research Question

This subsection addresses the following sub-research question:

> *How do activation-transfer size and compute distribution explain the latency and boundary-crossing behavior observed for the decomposition choices evaluated in RQ1.1 and RQ1.2?*

RQ1.3 is not an additional split benchmark. It is an explanatory synthesis stage that interprets the local findings already reported in RQ1.1 and RQ1.2 using two structural properties of the partition: the size of the intermediate activation tensor transferred at the boundary, and the distribution of computation across the two resulting services. To support this interpretation, RQ1.3 draws on the split benchmark measurements from RQ1.1 and RQ1.2 as primary evidence and uses an independently conducted internal timing experiment on the monolithic ResNet-18 model as supporting evidence for the compute-distribution explanation.

The literature framing for activation-transfer size, compute distribution, and the ResNet-18 stage structure used as the basis for this synthesis is discussed in sections 3.1 and 3.2.

5.3.2 Analytical Setup

RQ1.3 draws on three bodies of evidence:

1. **Split benchmark measurements from RQ1.1.** These include the mean end-to-end latencies, boundary-crossing intervals, and activation-transfer sizes for the five coarse conditions (monolithic plus four stage-level splits). The relevant data are summarised in table 5.2 and the overhead analysis in section 5.1.5.
2. **Split benchmark measurements from RQ1.2.** These include the mean end-to-end latencies, boundary-crossing intervals, and activation-transfer sizes for the four refined conditions (monolithic plus three refined splits within the carried-forward region). The relevant data are summarised in table 5.4 and the overhead analysis in section 5.2.5.
3. **Internal timing experiment on monolithic ResNet-18.** A separate instrumented forward-pass profiling experiment was conducted on the same monolithic model under a closely matched single-threaded CPU-stabilised environment, frozen as `internal_timing_resnet18_20260515_113614`. This experiment measured per-stage, per-block, and selected internal operation timings for a single-threaded CPU forward pass, averaged over 200 measured iterations preceded by 50 warmup iterations, using 10 rounds and the same randomisation seed. Functional equivalence was verified (maximum absolute difference of 0.0 against the uninstrumented model in `validation.json`), and instrumentation overhead was measured at 0.708 ms (0.94% of total forward-pass time, per `instrumentation_overhead.json`), confirming that the profiling did not meaningfully distort the timing distribution.

Table 5.5: RQ1.3 supporting internal timing configuration. Source: `config.yaml` and `instrumentation_overhead.json` in `internal_timing_resnet18_20260515_113614`.

Setting	Value
Model	ResNet-18 (pretrained)
Device	CPU
Precision	FP32
Profiling mode	Monolithic internal timing (supporting evidence)
Timing scope	Stage-, block-, and selected operation-level timing
Rounds	10
Warmup iterations	50
Measured iterations	200
Validation inputs	5
Validation tolerance	1.0×10^{-6}
CPU stabilisation	Single-threaded, single-core pinned execution
Instrumentation overhead	0.708 ms (0.94%)

> This experiment is used as supporting evidence to characterise the internal compute distribution of ResNet-18; it is not a split benchmark and does not introduce new split conditions.

No new split benchmark was conducted for RQ1.3. The explanatory analysis is based entirely on measurements already reported in RQ1.1 and RQ1.2, supplemented by the internal timing experiment described above. To keep this supporting evidence auditable without turning RQ1.3 into a separate benchmark stream, tables 5.5 to 5.7 and fig. 5.2 provide a compact summary of the internal timing configuration and the specific internal timing results used in the explanatory analysis.

5.3.3 Explanatory Findings

The findings are organised around four explanatory observations, each building on the previous one.

A. Activation-transfer burden explains much of the coarse latency pattern. In RQ1.1, the boundary-crossing interval decreased monotonically from the earliest split to the latest: 53.81 ms after `layer1` (784 KiB activation), 40.85 ms after `layer2` (392 KiB), 26.06 ms after `layer3` (196 KiB), and 2.22 ms after `layer4` (98 KiB). This pattern closely tracks the reduction in activation-transfer size as the split point moves deeper into the network. Larger intermediate tensors require more serialisation, transmission, and deserialisation time over the gRPC boundary, and this cost is directly reflected in the boundary-crossing measurements.

This observation is consistent with the broader partitioning literature, which identifies activation size as a primary driver of communication overhead in distributed inference [7, 15, 30, 31, 33]. At the coarse level, the monotonic relationship between activation

Table 5.6: RQ1.3 supporting internal timing summary used in the explanatory analysis. Source: `summary.csv` in `internal_timing_resnet18_20260515_113614`. Classification head is the sum of `avgpool` and `fc`. The convolution and batch-norm rows are aggregated over all matching units at level L3 in the same summary file.

Unit / Metric	Mean (ms)	% of model
Model total	75.377	100.00%
Stem	12.172	16.15%
Layer1	13.132	17.42%
Layer2	12.172	16.15%
Layer3	13.961	18.52%
Layer4	23.576	31.28%
Layer3.1 block	7.475	9.92%
Classification head	0.330	0.44%
Convolution ops (L3 agg.)	61.830	82.0%
BatchNorm ops (L3 agg.)	2.580	3.4%

Table 5.7: RQ1.3 heaviest internal compute units from the supporting timing analysis. Source: `summary.csv` in `internal_timing_resnet18_20260515_113614`, sorted by `mean_ms` at level L3.

Unit	Mean (ms)	% of model
`stem.maxpool`	7.267	9.64%
`layer4.1.conv1`	6.157	8.17%
`layer4.0.conv2`	6.153	8.16%
`layer4.1.conv2`	6.103	8.10%
`layer4.0.conv1`	4.002	5.31%
`layer3.0.conv2`	3.669	4.87%
`layer3.1.conv1`	3.625	4.81%
`stem.conv1`	3.615	4.80%

size and boundary-crossing interval shows that transfer burden is a dominant explanatory factor for the boundary-crossing cost gradient observed across the four RQ1.1 split points.

B. Activation size alone does not determine suitability. Although `split_after_layer4` achieved the lowest boundary-crossing interval in RQ1.1 and one of the lowest raw split mean latencies (79.58 ms), it was classified as compute-degenerate because the second service executed only average pooling and the final linear layer — a residual computation segment well under 1 ms. This means that minimising activation-transfer size to its smallest value does not, by itself, produce a suitable microservice decomposition. A boundary that pushes nearly all computation to one side leaves the second service structurally empty, which defeats the purpose of decomposition.

The internal timing experiment provides quantitative support for this observation.

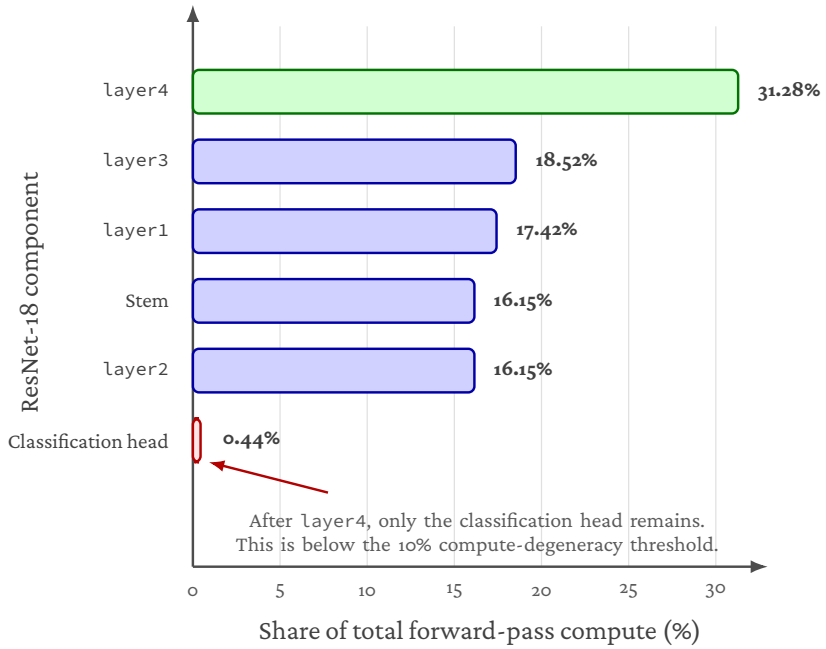


Figure 5.2: Stage-level compute distribution from the RQ1.3 internal timing analysis. The final classification head contributes less than 1% of total forward-pass compute, while `layer4` accounts for the largest stage-level share. This explains why `split_after_layer4` is raw-fast but compute-degenerate: almost all meaningful model computation remains before the split.

As shown in fig. 5.2, the monolithic forward pass took approximately 75.38 ms on average, of which `layer4` alone accounted for 31.28% (approximately 23.58 ms). More importantly, the computation remaining after a `layer4` split — average pooling plus the final linear layer — accounts for only about 0.33 ms in total, or roughly 0.44% of the full forward pass. This means that a split after `layer4` places approximately 99.6% of the total computation on the first service and leaves only about 0.4% on the second. By contrast, a split after `layer3` places approximately 68.3% of compute on the first service and 31.7% on the second when `layer4`, pooling, and classification are counted together, producing a substantially more balanced distribution.

The supporting timing analysis also shows that compute is concentrated in a small number of heavy internal units rather than being spread uniformly. As shown in table 5.7, the heaviest units are dominated by convolutional operations in later network regions, especially `layer4`. This makes a very late split structurally problematic: it transfers a small activation tensor, but it also leaves the remote side with almost no substantial computation.

This distinction between raw-fastest and structurally suitable is consistent with literature that treats compute distribution as a first-class partition criterion alongside communication cost [14, 18, 20, 21].

C. The RQ1.2 controlled comparison clarifies the role of compute placement. The most precise supporting evidence for the role of compute distribution comes from the RQ1.2 comparison between `split_after_layer3_block0` and `split_after_layer3`. Both conditions transfer an intermediate tensor of approximately 196 KiB. If activation-transfer size were the sole determinant of boundary-crossing cost, these two conditions should exhibit similar boundary-crossing intervals and similar end-to-end latencies. They do not.

`split_after_layer3_block0` exhibited a boundary-crossing interval of 33.97 ms, while `split_after_layer3` exhibited a boundary-crossing interval of 26.16 ms — a difference of approximately 7.8 ms despite identical activation size. The internal timing data helps explain this gap. The `layer3.1` block (the second residual block of `layer3`) accounts for approximately 9.92% of total monolithic compute. The difference is consistent with the amount of computation that remains on the second service. When the split is placed after `layer3.0`, Service B executes `layer3.1` in addition to `layer4`, pooling, and classification. When the split is placed after the full `layer3`, that computation remains on Service A. The internal timing report shows that `layer3.1` accounts for approximately 7.48 ms of monolithic compute, which closely matches the observed 7.81 ms difference in boundary-crossing interval between the two refined conditions. This provides strong supporting evidence that the boundary-crossing difference is mainly associated with compute placement on the remote side rather than with activation-transfer size, which is identical in both cases.

This comparison also shows why a purely tensor-size-based explanation is incomplete. The two refined conditions share the same activation size, but the internal work performed after the boundary differs materially. In that sense, RQ1.2 holds activation size constant while making compute placement the main observed difference within the carried-forward region.

This observation is consistent with the principle established in the partitioning literature that compute distribution must be considered alongside communication when interpreting end-to-end split behavior and boundary-crossing measurements [14, 18, 28].

D. Internal timing confirms that compute is unevenly distributed within ResNet-18. The internal timing experiment provides a structural explanation for why different boundaries within the same coarse region yield different performance. As visualised in fig. 5.2, the per-stage compute shares are approximately: stem 16.15%, `layer1` 17.42%, `layer2` 16.15%, `layer3` 18.52%, and `layer4` 31.28%. Aggregating across the level-3 operation rows in table 5.6, convolution operations account for approximately 82% of total compute, while batch normalisation contributes approximately 3% and individual ReLU or residual-addition operations are negligible.

The operation-level support data adds a more concrete view of this distribution. As shown in table 5.7, the heaviest internal units are predominantly convolutional operations, with the largest shares concentrated in `layer4` and the late `layer3` region. By contrast, batch normalisation, residual addition, and ReLU operations are individually small and do not serve as meaningful compute anchors by themselves. This confirms that ResNet-18 is not only uneven at the stage level, but also internally dominated by a

relatively small number of heavy convolutional units.

Two aspects of this distribution are relevant to the RQ1.1 and RQ1.2 findings. First, `layer4` is by far the most compute-intensive stage, which explains why a split after `layer4` is structurally degenerate: it assigns the dominant compute block entirely to the first service. Second, the per-block decomposition within `layer3` reveals that `layer3.0` and `layer3.1` do not contribute equally; the difference in their execution times contributes to the boundary-crossing asymmetry observed between `split_after_layer3_block0` and `split_after_layer3`. This structural unevenness is consistent with the design of ResNet-18, where each successive stage doubles the channel count and halves the spatial resolution, producing progressively heavier convolution operations in later stages [11].

It should be noted that the internal timing experiment was conducted on the monolithic model and therefore measures local per-layer execution cost without the serialisation and communication overhead present in the split benchmarks. The timing values are used here to explain the compute-distribution dimension; the actual boundary-crossing cost is a composite of both communication and compute-related factors as measured in the split benchmarks themselves.

The interpretation of these findings and the answer to RQ1.3 are presented in section 6.1.3.

5.4 RQ1.4 Multi-Microservice Kubernetes Inference Chain

5.4.1 Research Question

> *How does increasing the number of microservices affect end-to-end inference cost when ResNet-18 is decomposed into synchronously chained coarse architectural stages under in-cluster Kubernetes execution?*

RQ1.4 moves from the local split-selection and explanatory stages of RQ1.1–RQ1.3 to the final thesis-facing local Kubernetes benchmark for the coarse-stage ResNet-18 chain family. Rather than selecting a boundary through carry-forward, this stage benchmarks a predefined set of five in-cluster conditions from `monolithic_k8s_1svc` to `chain_5svc` to measure how end-to-end latency changes as synchronous service boundaries are added under controlled local Kubernetes execution.

The literature framing for multi-boundary chain cost and Kubernetes-level deployment overhead is discussed in sections 3.1 and 3.3.

5.4.2 Experimental Setup

The thesis-facing RQ1.4 benchmark used the frozen controlled local Kubernetes run `rq1_4_fully_controlled_20260515_141036`. The environment snapshot (`merged_results/environment.j` records an Ubuntu/Linux runtime (`Linux-6.17.0-23-generic-x86_64-with-glibc2.41`), Python 3.10.20, and PyTorch 2.11.0+cpu. Deployment metadata (`merged_results/deployment_metadata.json`) shows that the benchmark ran in namespace `rq14` on a single-node `kind` cluster (`kind-thesis-rq14`); for every condition, all service pods were scheduled on the same node, `thesis-rq14-control-plane`. The container image is the mutable local tag `thesis-inference:latest` with pull policy `IfNotPresent`, as documented in the manifest's reproducibility note (section 4.8). The evaluated model was pretrained ResNet-18 with CPU-only FP32 execution and input shape $1 \times 3 \times 224 \times 224$.

CPU stabilisation controls were requested for the benchmark processes and applied where the runtime allowed. Threading controls for PyTorch, OpenMP, MKL, and OpenBLAS were all requested at one thread and observed at one thread. CPU affinity to one physical core with SMT excluded was requested and applied; the observed affinity set was core 0. Process priority `nice -5` was requested and applied. Governor and turbo controls were also requested, but writes to `/sys` failed because the containerised environment exposed those paths as read-only. The detected governor was already `performance`, and turbo was already disabled. ASLR remained at the detected value 2 and was not modified.

Methodological validity was established before timing. For all five conditions, local segment validation against the monolithic model and live gRPC chain validation against the deployed services both passed with `max_abs_diff` $= 0.0$, `atol` $= 1 \times 10^{-5}$, and `num_inputs` $= 5$.

5.4.3 Conditions

Five conditions were evaluated in a rolling orchestration sweep:

Table 5.8: RQ1.4 benchmark configuration

Setting	Value
Runtime	Local Ubuntu/Linux Kubernetes benchmark
Cluster	Single-node `kind` cluster (`kind-thesis-rq14`)
Namespace	`rq14`
Conditions	Five predefined configurations from `monolithic_k8s_1svc` to `chain_5svc`
Execution mode	Local in-cluster rolling orchestration over the predefined condition set
Model	ResNet-18 (pretrained)
Device	CPU
Precision	FP32
Input shape	$1 \times 3 \times 224 \times 224$
Rounds	5
Warmup iterations	50
Measured iterations	200
Cooldown between conditions	5 seconds
Random seed	42
Communication	gRPC (in-cluster Kubernetes)
gRPC port	50051
Maximum message size	16 MiB ($16{,}777{,}216$ bytes)
Service pod CPU request / limit	1 core / 1 core
Service pod memory request / limit	512 MiB / 512 MiB
Client CPU request / limit	1 core / 1 core
Client memory request / limit	1 GiB / 1 GiB
Image pull policy	`IfNotPresent`
Carry-forward	Not applicable; RQ1.4 benchmarks a predefined configuration set
Parity validation	Local segment validation and live gRPC chain validation; `num_inputs` $= 5$, `atol` $= 1 \times 10^{-5}$
Warmup calibration	Trailing window $= 10$, CV threshold $= 0.02$

The byte totals reported below are the frozen runner's `total_activation_bytes_mean` values converted with 1024 bytes/KiB. This runner metric sums the input tensor and the tensors recorded across the service calls, so it is a per-request recorded tensor-byte total rather than the boundary-only activation-transfer size used in the local split-selection sections.

- **`monolithic_k8s_1svc`**: Full ResNet-18 inference in a single Kubernetes service. This is the Kubernetes-native monolithic baseline.

- **chain_2svc**: Two services with a split after `layer2`. Recorded tensor bytes per request: 980 KiB.
- **chain_3svc**: Three services with splits after `layer1` and `layer3`. Recorded tensor bytes per request: 1568 KiB.
- **chain_4svc**: Four services with splits after `layer1`, `layer2`, and `layer3`. Recorded tensor bytes per request: 1960 KiB.
- **chain_5svc**: Five services with splits after `layer1`, `layer2`, `layer3`, and `layer4`. Recorded tensor bytes per request: 2058 KiB.

Carry-forward is not applicable in RQ1.4. The frozen `carry_forward.json` states that carry-forward is defined only for the two-service split-selection experiments (RQ1.1 and RQ1.2), whereas RQ1.4 benchmarks this predefined Kubernetes chain set.

5.4.4 Results

Table 5.9: RQ1.4 summary of mean latency and overhead vs. Kubernetes monolithic baseline. Source: `merged_results/condition_summaries.csv` in `rq1_4_fully_controlled_20260515_141036`.

Condition	Mean (ms)	Median (ms)	Std (ms)	p95 (ms)	Overhead vs. baseline
monolithic_k8s_1svc	81.066	79.506	4.017	88.250	—
chain_2svc	83.387	81.803	3.525	90.189	+2.321 / +2.9%
chain_3svc	87.728	85.775	4.233	95.269	+6.662 / +8.2%
chain_4svc	90.572	88.675	3.929	97.713	+9.506 / +11.7%
chain_5svc	92.535	90.456	3.965	99.491	+11.469 / +14.1%

Figure 5.3 visualises the same latency trend reported in table 5.9. By `chain_4svc`, the mean-latency overhead is 9.506 ms out of the final `chain_5svc` overhead of 11.469 ms, so most of the measured increase has already occurred before the five-service configuration.

Table 5.10: RQ1.4 overhead and non-compute/platform cost breakdown. Recorded tensor-byte totals are taken from `total_activation_bytes_mean` in `merged_results/condition_summaries.csv` and converted to KiB (1024 bytes/KiB); the metric includes the input tensor. The platform column is `non_compute_overhead_ms_mean`.

Condition	Overhead (ms)	Overhead (%)	Tensor Bytes (KiB)	Non-Compute / Platform (ms)
chain_2svc	2.321	2.9	980	6.215
chain_3svc	6.662	8.2	1568	9.435
chain_4svc	9.506	11.7	1960	12.088
chain_5svc	11.469	14.1	2058	13.633

Methodological note. Effect sizes are computed from pooled per-iteration data ($N = 1{,}000$ iterations per condition). With large N, Mann-Whitney p-values are near-zero for any non-trivial difference and should not be over-interpreted as evidence of strong practical significance. Cohen's d provides a more informative measure of effect magnitude. Cross-round consistency and parity validation are the primary indicators of result stability and methodological validity.

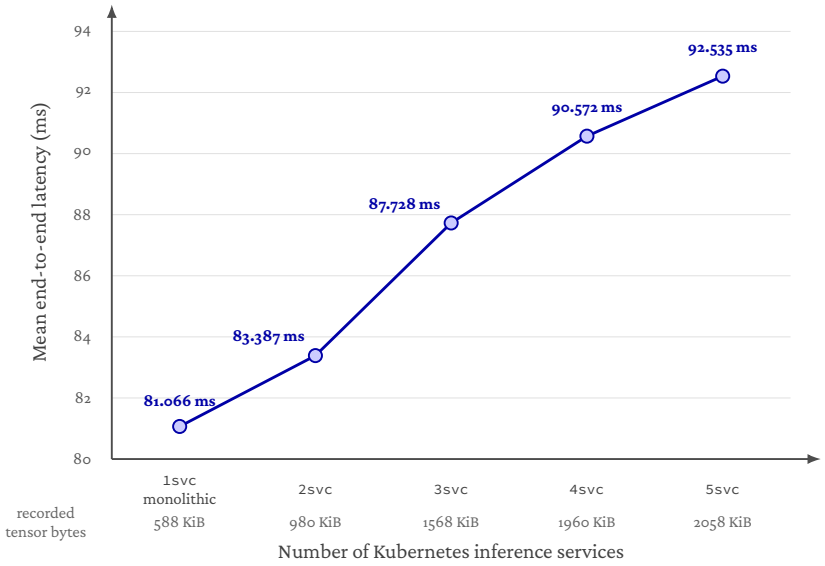


Figure 5.3: Mean end-to-end latency for the RQ1.4 local Kubernetes chain family (`rq1_4_fully_controlled_20260515_141036`). Latency increases as the predefined service-chain configurations move from one to five services, but the adjacent differences are descriptive comparisons across this predefined family rather than isolated causal estimates for a single added boundary.

Table 5.11: RQ1.4 effect sizes relative to monolithic baseline. Source: `merged_results/effect_sizes.csv` in `rq1_4_fully_controlled_20260515_141036`.

Condition	**Cohen's d**	**Interpretation**	**Mann-Whitney p**
`chain_2svc`	0.614	medium	7.66×10^{-77}
`chain_3svc`	1.614	large	4.02×10^{-181}
`chain_4svc`	2.392	large	3.23×10^{-276}
`chain_5svc`	2.874	large	1.75×10^{-301}

The frozen controlled local Kubernetes run was stable across rounds. Round CVs remained low for all five conditions: 0.0085 for `monolithic_k8s_1svc`, 0.0099 for `chain_2svc`, 0.0161 for `chain_3svc`, 0.0097 for `chain_4svc`, and 0.0074 for `chain_5svc`. Together with the parity checks above, this supports treating the reported condition means as representative of the local in-cluster benchmark.

The interpretation of these results and the answer to RQ1.4 are presented in section 6.2.1.

Table 5.12: RQ1.4 cross-round consistency. Source: `merged_results/round_consistency.json` in `rq1_4_fully_controlled_20260515_141036`.

Condition	Grand Mean (ms)	Round Std (ms)	Round Range (ms)	Round CV
monolithic_k8s_1svc	81.066	0.6864	1.8009	0.0085
chain_2svc	83.387	0.8288	2.0200	0.0099
chain_3svc	87.728	1.4109	3.8162	0.0161
chain_4svc	90.572	0.8826	2.0073	0.0097
chain_5svc	92.535	0.6820	1.6288	0.0074

5.5 RQ1.5 Azure Kubernetes Service Transfer Validation

5.5.1 Research Question

> *Does the local Kubernetes latency pattern observed for the coarse-stage ResNet-18 microservice chain transfer to Azure Kubernetes Service, and how do the absolute and relative costs change under managed cloud execution?*

RQ1.5 moves the predefined RQ1.4 chain family from local `kind` Kubernetes to Azure Kubernetes Service (AKS). The purpose is not to select a new decomposition, but to test whether the architectural ordering observed in local Kubernetes remains credible when the same synchronously chained inference workload is executed on managed cloud infrastructure. The stage therefore compares absolute latency, relative overhead, and condition ordering between the frozen RQ1.4 local Kubernetes run and the frozen RQ1.5 AKS run.

The literature framing for transfer-validation between local Kubernetes and managed cloud execution is discussed in section 3.3: managed Kubernetes, CNI, and cloud-network studies motivate treating local-to-cloud transfer as a directional comparison rather than a millisecond-identical reproduction [6, 8, 16], while benchmarking guidance motivates reporting ranking, tails, and cross-round stability alongside means [13].

5.5.2 Experimental Setup

The thesis-facing RQ1.5 benchmark used the frozen controlled AKS transfer-validation benchmark `rq1_5_fully_controlled_20260515_145954`. The deployment ran in namespace `rq15` on a single-node AKS cluster, with `Standard_D8s_v3` nodes (per `deployment_metadata.json`, `node_instance_type`) in the `swedencentral` Azure region. Deployment metadata confirms that pod colocation was enforced: for every condition, the benchmark client and all service pods were scheduled on the same AKS node (`aks-rq15pool-42653082-vmss000000`). All AKS conditions used the same pinned container image digest: `sha256:d292aef407ee8a15da2bdaa680de1cc24b0975e7de81d27e1fc3a7d128b8736b`.

Threading, affinity, and priority controls followed the same stabilisation intent as the preceding stages and were applied where the managed AKS runtime allowed. Threading controls for PyTorch, OpenMP, MKL, and OpenBLAS were requested and observed at one thread. CPU affinity was applied to one physical core with SMT excluded; the observed affinity set was core 0. Process priority `nice -5` was requested but could not be applied due to insufficient privileges, so the benchmark ran at `nice=0`. CPU governor and turbo controls were deliberately disabled in the AKS configuration because the relevant sysfs interfaces were unavailable. These constraints are treated as realistic cloud-environment limitations of managed or virtualised execution, not as architectural effects of the microservice decomposition.

Before timing, both local chain-segment validation and live gRPC validation passed for every condition with `max_abs_diff` $= 0.0$ and `atol` $= 1 \times 10^{-5}$. This supports functional equivalence across the monolithic and chained AKS deployments.

5.5.3 Conditions

RQ1.5 reused the same predefined chain family as RQ1.4:

- **`monolithic_k8s_1svc`**: Full ResNet-18 inference in a single Kubernetes service.
- **`chain_2svc`**: Two services with a split after `layer2`. Recorded tensor bytes per request: 980 KiB.
- **`chain_3svc`**: Three services with splits after `layer1` and `layer3`. Recorded tensor bytes per request: 1568 KiB.
- **`chain_4svc`**: Four services with splits after `layer1`, `layer2`, and `layer3`. Recorded tensor bytes per request: 1960 KiB.
- **`chain_5svc`**: Five services with splits after `layer1`, `layer2`, `layer3`, and `layer4`. Recorded tensor bytes per request: 2058 KiB.

Carry-forward is not applicable in this stage. The experiment evaluates a fixed deployment family in AKS rather than choosing a new split point. As in RQ1.4, the recorded tensor-byte totals are taken from the runner metadata and include the input tensor as well as hop activations; they are therefore used as a consistent input-inclusive traffic proxy, not as a boundary-only activation-transfer size.

5.5.4 Results

The transfer comparison was derived directly from the frozen RQ1.4 and RQ1.5 summary CSV artifacts, because the generated RQ1.5 report does not itself embed the cross-stage comparison.

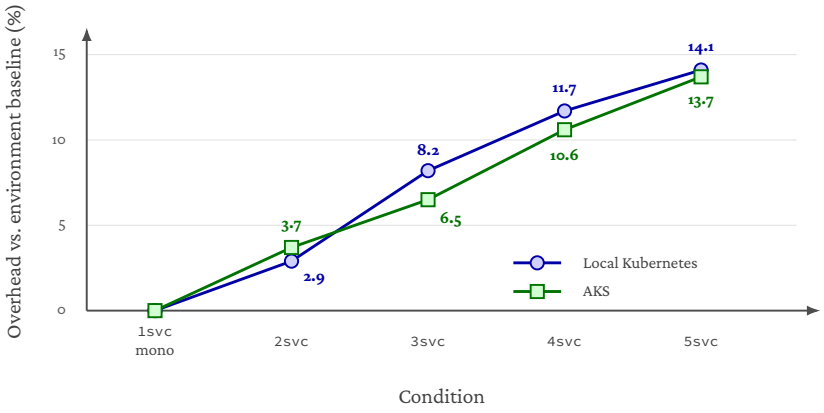


Figure 5.4: RQ1.5 normalised overhead transfer comparison between local Kubernetes and AKS. Overheads are computed relative to each environment's own monolithic Kubernetes baseline, so the figure visualises directional transfer of the service-count trend rather than raw latency equivalence.

Figure 5.4 visualises the transfer comparison in table 5.16. The overhead trend is similar across environments even though the absolute millisecond values differ.

The rank ordering is identical between the frozen local Kubernetes and AKS runs: `monolithic_k8s_1svc`, `chain_2svc`, `chain_3svc`, `chain_4svc`, and `chain_5svc`. Absolute latency is lower in the AKS run for every condition, but the within-environment ordering and the direction of the overhead trend are preserved.

The AKS run is stable across measured rounds for all chained conditions, with round CVs of 0.0019–0.0061. The monolithic baseline shows a slightly higher round CV of 0.0117. This pattern is compatible with managed-cloud variability on the `Standard_D8s_v3` VM, but the run does not isolate a single cause; in this frozen measurement, the chained conditions show lower round-level CVs than the monolithic baseline. Warmup calibration indicated that each run reached a stable trailing-window CV at least once before measurement, while the cross-round CVs of the measured iterations provide the stronger stability evidence for the final reported means. As in the previous stage, effect sizes and Mann-Whitney tests were computed from pooled per-iteration data and are treated as supplementary because $N = 1{,}000$ per condition makes very small differences statistically significant. Cross-round stability, parity validation, and preserved ordering are the primary evidence for this stage.

The interpretation of these results and the answer to RQ1.5 are presented in section 6.2.2.

5.6 RQ1.5b AKS Multi-Node Sensitivity

5.6.1 Research Question

> *When every chain hop is forced across a node boundary, how much additional end-to-end latency does the inter-node network path add, relative to the same-node RQ1.5 baseline?*

RQ1.5b is a sensitivity stage. It does not replace RQ1.5; the frozen single-node AKS results in section 5.5 remain the thesis-facing primary numbers. The role of RQ1.5b is to quantify, in milliseconds, the inter-node penalty that the single-node design intentionally hides, so that the single-node deployment threat to validity becomes a bounded, measured cost rather than an unspecified one.

5.6.2 Experimental Setup

The thesis-facing RQ1.5b benchmark used the frozen multi-node sensitivity run `rq1_5b_multinode_20260515_152628`. The deployment ran in namespace `rq15b` on a multi-node AKS cluster in the `swedencentral` Azure region. All worker nodes were `Standard_D8s_v3` (per `node_instance_type` in `deployment_metadata.json`) and shared one node pool, so the only intentional difference from RQ1.5 is the pod-to-node assignment.

Multi-node placement was enforced by Kubernetes pod anti-affinity. Each service pod of a given condition repels every other service pod of that condition keyed on `kubernetes.io/hostname`, and the benchmark client repels every pod with `workload-role: service`. The combined rule guarantees that for the `chain_5svc` condition all

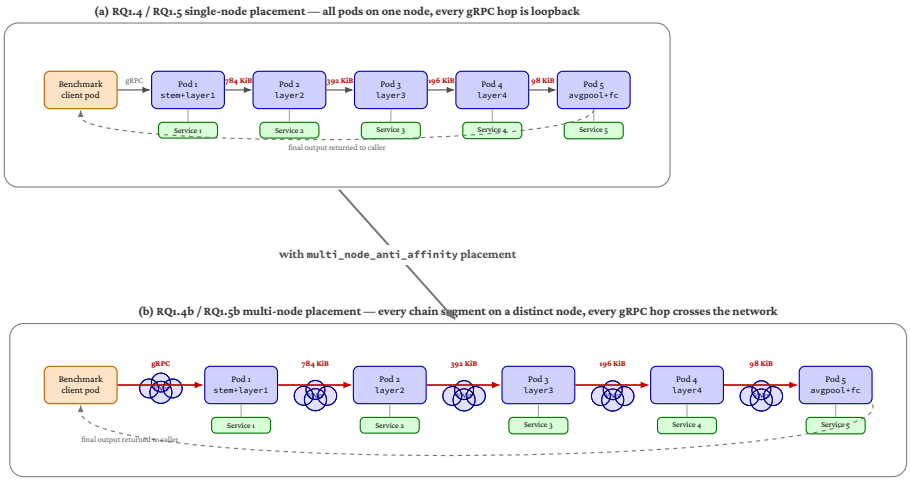


Figure 5.5: Same chain, two topologies. (a) RQ1.4 / RQ1.5 single-node placement: all six pods on one node, every gRPC hop is loopback. (b) RQ1.4b / RQ1.5b multi-node placement: each chain segment lands on a distinct cluster node and the client lands on a sixth dedicated node, so every gRPC hop crosses the Azure VNet. The chain itself is identical between the two panels; only the pod-to-node assignment changes.

five service segments land on distinct nodes and the client lands on a sixth dedicated node, so every gRPC hop in the inference path crosses the Azure VNet rather than staying loopback. Compliance was verified after deployment by reading observed `nodeName` values from the Kubernetes API. For every condition, the recorded service nodes were pairwise disjoint and the client node was distinct from every service node; no condition was promoted to the merge phase without this check passing. Functional parity was independently verified: `max_abs_diff` $= 0.0$ at `atol` $= 1 \times 10^{-5}$ for all five conditions in both the local segment validation and the live gRPC chain validation (`num_inputs` $= 5$).

The shared benchmark contract is otherwise unchanged from RQ1.5: ResNet-18 with pretrained weights, CPU FP32 inference, the same five chain conditions, the same five-round / 200-iteration / 50-warmup discipline, the same Guaranteed-QoS pod profile (1 vCPU + 1 GiB per pod), and the same image used in RQ1.5 (`sha256:d292aef407ee8a15da2b daa680de1cc24b0975e7de81d27e1fc3a7d128b8736b`).

5.6.3 Conditions

RQ1.5b reuses the same predefined chain family as RQ1.4 and RQ1.5 (`monolithic_k8s_1svc`, `chain_2svc`, `chain_3svc`, `chain_4svc`, `chain_5svc`). The full family is included rather than a single representative because the per-condition delta is the headline metric: a single point cannot show whether the inter-node penalty grows with hop count, dominates one specific boundary, or stays roughly flat. Five points support the direct comparison shown in table 5.19.

5.6.4 Results

The within-environment ordering is preserved (`monolithic_k8s_1svc` < `chain_2svc` < `chain_3svc` < `chain_4svc` < `chain_`
and the bounded-non-linearity finding from RQ1.4 reproduces under multi-node execution: the `chain_4svc` → `chain_5svc` increment is 2.502 ms, broadly comparable to the preceding 4.562 ms step. Cross-round CVs are 0.0036–0.0127 across all chained conditions, consistent with placing each pod on its own dedicated VM rather than sharing a single host.

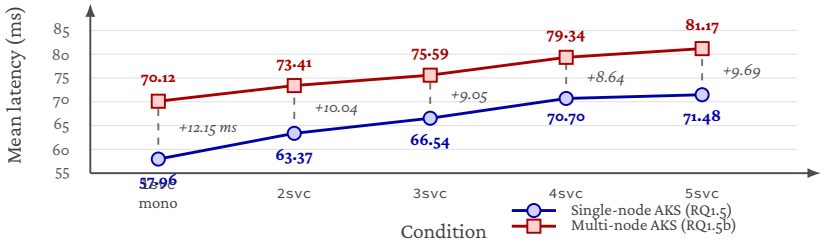


Figure 5.6: RQ1.5b multi-node penalty per condition. Same SKU (`Standard_D8s_v3`), same region (`swedencentral`); the only difference is single-node colocation versus `multi_node_anti_affinity` placement. Per-condition deltas range from 2.131 ms (`chain_3svc`) to 4.086 ms (`chain_5svc`), and the monolithic-baseline penalty is 2.306 ms.

Table 5.19 and fig. 5.6 summarise the per-condition multi-node penalty. The penalty is positive across all conditions and grows with chain depth. The monolithic baseline alone gains 2.306 ms relative to the same single-node monolithic, because the client now ships a 600 KiB input tensor across the Azure VNet rather than across a loopback socket. Adding chain hops with smaller intermediate activations (784, 392, 196, 98 KiB) yields per-condition deltas between 2.131 ms and 4.086 ms, with the deepest `chain_5svc` configuration carrying the largest absolute penalty. The within-environment monotonic ordering is preserved.

The interpretation of these results and the answer to RQ1.5b are presented in section 6.2.3.

Table 5.13: RQ1.5 AKS benchmark configuration. Source: `merged_results/config.yaml` and `deployment_metadata.json` in `rq1_5_fully_controlled_20260515_145954`.

Setting	**Value**
Runtime	Azure Kubernetes Service benchmark
Cluster	Single-node AKS cluster `thesis-rq15`
Node type	`Standard_D8s_v3`
Region	`swedencentral`
Namespace	`rq15`
Conditions	Five predefined configurations from `monolithic_k8s_1svc` to `chain_5svc`
Execution mode	In-cluster rolling orchestration with teardown between conditions
Model	ResNet-18 (pretrained)
Device	CPU
Precision	FP32
Input shape	$1 \times 3 \times 224 \times 224$
Rounds	5
Warmup iterations	50
Measured iterations	200
Cooldown between conditions	5 seconds
Random seed	42
Communication	gRPC (in-cluster Kubernetes)
gRPC port	50051
Maximum message size	16 MiB
Service pod CPU request / limit	1 core / 1 core
Service pod memory request / limit	1 GiB / 1 GiB
Client CPU request / limit	1 core / 1 core
Client memory request / limit	1 GiB / 1 GiB
Pod QoS	`Guaranteed` for service pods
Image pull policy	`Always`
Carry-forward	Not applicable; RQ1.5 evaluates a predefined chain family
Parity validation	Local segment validation and live gRPC chain validation; `num_inputs` $= 5$, `atol` $= 1 \times 10^{-5}$
Warmup calibration	Trailing window $= 10$, CV threshold $= 0.02$

Table 5.14: RQ1.5 AKS summary of mean latency and overhead vs. AKS monolithic baseline. Source: merged_results/condition_summaries.csv in rq1_5_fully_controlled_20260515_145954.

Condition	Mean (ms)	Median (ms)	Std (ms)	p95 (ms)	Overhead vs. baseline
monolithic_k8s_1svc	70.802	70.224	2.659	75.151	—
chain_2svc	73.437	73.124	1.831	76.602	+2.635 / +3.7%
chain_3svc	75.410	75.043	1.971	78.301	+4.608 / +6.5%
chain_4svc	78.294	77.923	2.190	81.698	+7.492 / +10.6%
chain_5svc	80.519	80.078	2.158	84.635	+9.717 / +13.7%

Table 5.15: RQ1.5 AKS overhead and inferred non-compute/platform cost breakdown. Source: merged_results/condition_summaries.csv in rq1_5_fully_controlled_20260515_145954; the platform column is non_compute_overhead_ms_mean.

Condition	Overhead (ms)	Overhead (%)	Recorded Tensor Bytes (KiB)	Inferred Non-Compute / Platform (ms)
chain_2svc	2.635	3.7	980	4.489
chain_3svc	4.608	6.5	1568	7.073
chain_4svc	7.492	10.6	1960	9.138
chain_5svc	9.717	13.7	2058	10.490

Table 5.16: RQ1.5 transfer comparison between local Kubernetes and AKS. Source: merged_results/condition_summaries.csv in rq1_4_fully_controlled_20260515_141036 and rq1_5_fully_controlled_20260515_145954.

Condition	Local mean (ms)	AKS mean (ms)	Local overhead	AKS overhead
monolithic_k8s_1svc	81.066	70.802	0.0%	0.0%
chain_2svc	83.387	73.437	2.9%	3.7%
chain_3svc	87.728	75.410	8.2%	6.5%
chain_4svc	90.572	78.294	11.7%	10.6%
chain_5svc	92.535	80.519	14.1%	13.7%

Table 5.17: RQ1.5 AKS cross-round consistency. Source: frozen RQ1.5 merged_results/round_consistency.json.

Condition	Grand Mean (ms)	Round Std (ms)	Round Range (ms)	Round CV
monolithic_k8s_1svc	70.802	0.8255	1.9646	0.0117
chain_2svc	73.437	0.1398	0.3183	0.0019
chain_3svc	75.410	0.1879	0.5187	0.0025
chain_4svc	78.294	0.1960	0.4629	0.0025
chain_5svc	80.519	0.4911	1.1659	0.0061

Table 5.18: RQ1.5b AKS multi-node summary of mean latency and overhead vs. AKS multi-node monolithic baseline. Source: `merged_results/condition_summaries.csv` in `rq1_5b_multinode_20260515_152628`.

Condition	Mean (ms)	Median (ms)	Std (ms)	p95 (ms)	Overhead vs. baseline
monolithic_k8s_1svc	73.108	72.403	3.707	79.598	—
chain_2svc	76.470	75.653	3.435	82.696	+3.362 / +4.6%
chain_3svc	77.541	76.848	2.786	83.041	+4.433 / +6.1%
chain_4svc	82.103	81.660	2.355	86.394	+8.995 / +12.3%
chain_5svc	84.605	84.131	2.497	89.186	+11.497 / +15.7%

Table 5.19: RQ1.5b multi-node penalty, condition-by-condition vs. frozen RQ1.5 single-node AKS. Source: `merged_results/condition_summaries.csv` in `rq1_5_fully_controlled_20260515_145954` and `rq1_5b_multinode_20260515_152628`; overhead points are with respect to each environment's own monolithic baseline (tables 5.14 and 5.18).

Condition	Single-node mean (ms)	Multi-node mean (ms)	Δ (ms)	Δ overhead (pp)
monolithic_k8s_1svc	70.802	73.108	+2.306	—
chain_2svc	73.437	76.470	+3.033	+0.9
chain_3svc	75.410	77.541	+2.131	−0.4
chain_4svc	78.294	82.103	+3.809	+1.7
chain_5svc	80.519	84.605	+4.086	+2.0

Table 5.20: RQ1.5b AKS multi-node cross-round consistency. Source: `merged_results/round_consistency.json` in `rq1_5b_multinode_20260515_152628`.

Condition	Grand Mean (ms)	Round Std (ms)	Round Range (ms)	Round CV
monolithic_k8s_1svc	73.108	0.6134	1.5076	0.0084
chain_2svc	76.470	0.9679	2.3363	0.0127
chain_3svc	77.541	0.4388	0.9236	0.0057
chain_4svc	82.103	0.3757	0.9645	0.0046
chain_5svc	84.605	0.3047	0.6512	0.0036

5.7 RQ2.1 Service Identity and Mutual TLS Hardening

5.7.1 Research Question

> *What performance and operational overhead is introduced when the selected AKS microservice inference path is hardened with service identity, mutual TLS, and inter-service authorization policy?*

RQ2.1 adds the first security-hardening layer to the Azure deployment path selected from RQ1. The primary thesis-facing topology is `chain_2svc`, because it was the lowest-overhead non-monolithic chain in the preceding Kubernetes and AKS stages. The additional `chain_5svc` run is used as a stress case to examine whether the same security mechanism becomes more costly when the number of protected service boundaries increases. The goal is not to evaluate trusted execution environments in this stage, but to measure the cost of communication-level hardening: service identity, mutual TLS, and service-account-based authorization for inter-service traffic.

The literature framing for service-mesh-based mTLS, identity, and authorisation policy is discussed in section 3.4.

5.7.2 Experimental Setup

The RQ2.1 benchmark used two frozen paired AKS artifact sets: a chain-2 main driver `rq2_1_paired_20260515_160012` and a chain-5 maximum-depth stress driver `rq2_1_paired_20260517_114303`. Both runs used the same AKS cluster (`thesis-rq15`), strict same-node placement, and an isolated system node pool so that control-plane pods did not share the benchmark node.

All RQ2.1 conditions used the same pinned container image digest, shown across two lines for typesetting:

```
sha256:d292aef407ee8a15da2bdaa680de1cc2
    4b0975e7de81d27e1fc3a7d128b8736b
```

The mTLS condition used the managed AKS Istio add-on (`asm-1-29`). Each paired pass executed both security conditions; per `paired_summary.md` the alternation was mTLS-first on passes 1, 3, 5 and plain-first on passes 2, 4 in both topologies, so each condition has five executions and 1,000 measured iterations in total.

The plain condition used Kubernetes service-to-service communication without mesh injection. The mTLS condition added one Istio proxy per inference service, service accounts for the protected services, workload-level peer-authentication objects, and authorization-policy objects. The benchmark client remained outside the mesh. Consequently, RQ2.1 measures hardening of the inter-service path inside the inference chain rather than external ingress hardening. For `chain_2svc`, the downstream service was protected by strict mTLS and an authorization policy that allowed only the expected upstream service account. For `chain_5svc`, the same pattern was applied to each downstream segment, producing four immediate-upstream authorization policies.

Security validation passed in both frozen artifact sets. The mTLS runs passed static manifest validation and runtime validation, including the expected peer-authentication

Table 5.21: RQ2.1 paired AKS security benchmark configuration. Sources: `merged/paired_summary.md` in the two artifacts named below.

Setting	Value
Primary topology	`chain_2svc`; split after `layer2`
Stress topology	`chain_5svc`; splits after `layer1`, `layer2`, `layer3`, and `layer4`
Main artifact	`rq2_1_paired_20260515_160012`
Stress artifact	`rq2_1_paired_20260517_114303`
Conditions	Plain AKS chain vs. managed-Istio mTLS chain
Mesh revision	`asm-1-29`
Cluster	AKS `thesis-rq15`; isolated benchmark and system node pools
Benchmark node type	`Standard_D8s_v3`
Model	ResNet-18 (pretrained)
Device / precision	CPU / FP32
Input shape	$1 \times 3 \times 224 \times 224$
Paired passes	5
Warmup iterations	50 per condition execution
Measured iterations	200 per condition execution; 1,000 per condition after merging
Random seed	42
Service resources	1 CPU / 1 GiB per inference service
Client resources	100m CPU request, 1 CPU limit, 1 GiB memory
Sidecar resources	100m CPU request, 500m CPU limit, 128 MiB memory request, 512 MiB memory limit
Placement	Strict same-node service placement on tainted benchmark node pool
Security validation	Static manifest validation, runtime policy validation, full-chain positive probe, and non-mesh direct-denial probe

and authorization-policy counts. The full-chain positive probe passed for every mTLS pass, and direct non-mesh probes were blocked for the protected downstream segments. Resource metrics were complete in both runs.

5.7.3 Results

For the primary `chain_2svc` topology, the mTLS/AuthZ hardening bundle increased mean latency from 75.557 ms to 79.045 ms. The mean overhead was therefore 3.488 ms, or 4.62%, and p95 latency increased by 3.623 ms. The inferred non-compute/platform component increased from 4.559 ms to 7.891 ms. This supports the interpretation that

Table 5.22: RQ2.1 latency results for plain and mTLS AKS chains. Sources: `merged/aggregated_results.md` in `rq2_1_paired_20260515_160012` (`chain_2svc`) and `rq2_1_paired_20260517_114303` (`chain_5svc`).

Topology	Condition	n	Mean (ms)	Median (ms)	p95 (ms)	Inferred non-compute/platform (ms)
chain_2svc	plain	1000	75.557	75.262	80.047	4.559
chain_2svc	mtls	1000	79.045	78.590	83.670	7.891
chain_5svc	plain	1000	82.302	81.818	87.502	10.801
chain_5svc	mtls	1000	91.139	90.823	96.485	19.949

most of the added latency is associated with the communication/platform path introduced by the mesh rather than with a change in neural-network computation.

For the additional `chain_5svc` stress topology, the same hardening bundle increased mean latency from 82.302 ms to 91.139 ms. The mean overhead was 8.837 ms, or 10.74%, and p95 latency increased by 8.983 ms. The inferred non-compute/platform component increased from 10.801 ms to 19.949 ms. The stress run therefore shows that the communication-hardening cost grows when the number of protected service boundaries grows, even though the main RQ2.1 answer remains anchored in `chain_2svc`.

Table 5.23: RQ2.1 paired-pass latency deltas. Sources: `merged/aggregated_results.md` (per-pass tables) in `rq2_1_paired_20260515_160012` and `rq2_1_paired_20260517_114303`. The pass-delta std is computed from the five per-pass deltas.

Topology	Mean delta (ms)	Mean delta (%)	Min–max pass delta (ms)	Pass-delta std (ms)	p95 delta (ms)
chain_2svc	3.488	4.62	2.223–5.230	1.064	3.623
chain_5svc	8.837	10.74	7.623–9.985	0.847	8.983

The paired-pass deltas in table 5.23 are consistent with the overall means. In `chain_2svc`, every paired pass shows a positive mTLS/AuthZ hardening overhead, with pass-level deltas between 2.223 ms and 5.230 ms. In `chain_5svc`, every paired pass is also positive, with deltas between 7.623 ms and 9.985 ms. The alternating execution order reduces the risk that the result is explained only by slow drift during the run.

Figure 5.7 visualises the paired latency deltas from table 5.23. The figure focuses on deltas rather than raw latency so that the two RQ2.1 findings remain visible: the selected two-service chain pays a modest overhead, while the deeper stress chain pays a larger overhead once four inter-service boundaries are protected.

Table 5.24: RQ2.1 `chain_2svc` ablation of the communication-hardening bundle. Source: `merged/rq2_1_ablation_summary.md` in `rq2_1_ablation_20260516_112348`. As documented in the evidence manifest, ablation deltas are internally comparable only and must not be spliced into the frozen two-condition paired result.

Step	Condition	Mean latency (ms)	Adjacent delta (ms)	Interpretation
C0	Plain Kubernetes, no mesh	69.901	–	Non-mesh baseline
C1	Mesh-default sidecars / auto-mTLS / no AuthZ	72.470	+2.569	Main mesh data-path cost
C2	Strict downstream mTLS / no AuthZ	72.784	+0.315	Small added enforcement cost
C3	Strict downstream mTLS plus AuthZ	72.740	-0.044	Within pass-level noise

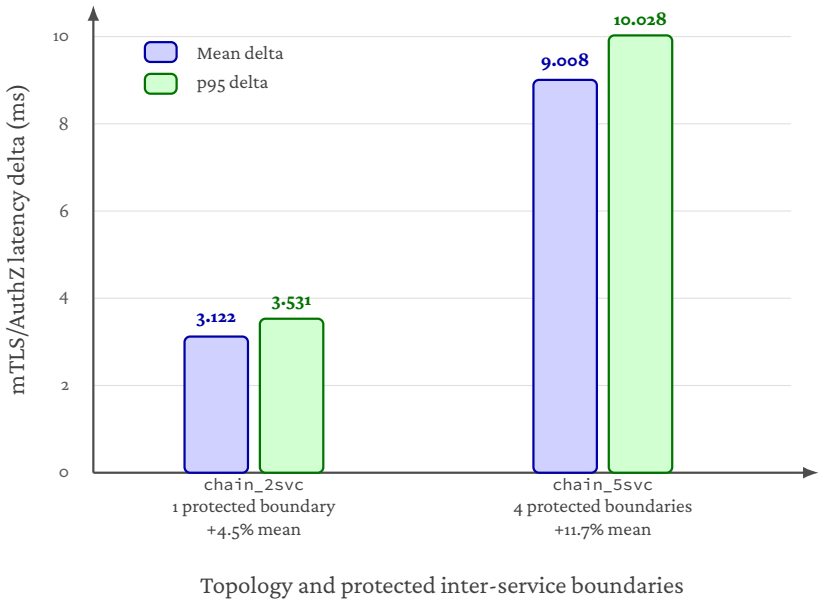


Figure 5.7: RQ2.1 mTLS/AuthZ latency overhead for the primary and stress AKS chains. Mean and p95 deltas are computed within the paired plain/mTLS design. The stress topology contains four protected inter-service boundaries, so the larger overhead visualises the depth sensitivity of sidecar-mediated communication hardening.

An additional `chain_2svc` ablation was run after the frozen paired benchmark to interpret the hardening bundle more precisely. This ablation is supporting evidence rather than a replacement for the frozen paired result; the evidence manifest states that the ablation campaign is internally comparable only and that its deltas must not be spliced into the frozen two-condition chain-2 / chain-5 numbers. With that caveat, the full hardened ablation condition reproduces the frozen result's order of magnitude: C3−C0 adds 2.840 ms, in the same range as the frozen paired `chain_2svc` overhead of 3.488 ms. As shown in table 5.24, most of the adjacent latency change occurs when moving from plain Kubernetes to the mesh-default condition. Strict mTLS enforcement over mesh-default traffic and the AuthorizationPolicy step are both small relative to pass-level variation. The correct interpretation is therefore that the deployed managed-Istio data path introduces most of the measured overhead, while strict mTLS and identity-based authorization mainly add enforcement semantics with little additional steady-state latency in this workload.

The mTLS condition also increased resource requests and deployment complexity. For `chain_2svc`, the mesh added two injected proxies, two service accounts, two peer-authentication objects, one authorization-policy object, 200m requested sidecar CPU, and 256 MiB requested sidecar memory. For `chain_5svc`, the mesh added five injected proxies, five service accounts, five peer-authentication objects, four authorization-policy

Table 5.25: RQ2.1 resource and operational overheads. Sources: `merged/paired_summary.md` in `rq2_1_paired_20260515_160012` (`chain_2svc`) and `rq2_1_paired_20260517_114303` (`chain_5svc`).

Topology	Sidecar CPU (mCPU)	Total pod CPU delta (mCPU)	Sidecar memory (MiB)	Total pod memory delta (MiB)	Schedule-to-ready delta (s)	Additional objects
chain_2svc	34.684	−56.813	82.125	93.819	6.30	5
chain_5svc	87.602	37.490	223.714	264.143	6.84	14

objects, 500m requested sidecar CPU, and 640 MiB requested sidecar memory. In both cases, the always-on Istio control-plane pods were present on the isolated system pool rather than the benchmark node. The CPU counter deltas should be read as window-level operational measurements rather than perfect causal attribution: in `chain_2svc`, measured application-container CPU was lower in the mTLS condition, which makes the total pod CPU delta negative despite a clearly positive sidecar contribution.

The interpretation of these results and the answer to RQ2.1 are presented in section 6.3.1.

5.8 RQ2.2 Confidential VM Execution Hardening

5.8.1 Research Question

> *What additional performance and operational overhead is introduced when VM-level confidential execution is added to the already communication-secured AKS inference chain, when the downstream protected service is placed on AMD SEV-SNP confidential nodes?*

RQ2.2 adds a second security-hardening layer to the Azure deployment path. RQ2.1 already established the communication-secured baseline: the selected `chain_2svc` inference pipeline runs on AKS with managed-Istio mTLS, service identity, and an authorisation policy on the protected downstream service. RQ2.2 keeps that communication-security contract fixed and changes the execution environment of the downstream service. The confidential condition places `service2` on an Azure confidential VM node pool backed by AMD SEV-SNP, while the standard condition uses matched non-confidential AMD VM nodes for both services. Per the canonical evidence manifest, the thesis-facing RQ2.2 scope is `service2_only`; placing both services on confidential nodes (full two-service TEE) is treated as future work and is not evaluated here. He et al. [12] motivate protecting the activations received by the downstream segment beyond the in-transit mTLS layer, although they evaluate a different (malicious remote-service) threat model than the SEV-SNP infrastructure-level protection measured here.

The purpose of this stage is therefore incremental. It does not ask whether mTLS is useful; that was answered in RQ2.1. Instead, it asks what extra cost is paid when runtime protection against a stronger infrastructure threat model is added on top of the already secured service-to-service path.

The literature framing for AMD SEV-SNP, VM-level confidential computing, and its interaction with sidecar-based service meshes is discussed in section 3.4.

5.8.2 Experimental Setup

RQ2.2 used the frozen rolling AKS artifact `rq2_2_confidential_20260516_135726`. The run preserves the RQ2.1 managed-Istio mTLS and authorisation semantics, and collects newly paired standard baselines on `Standard_D8as_v5` nodes rather than reusing the RQ2.1 numbers. The application image is the same ACR digest used in RQ1.5 and RQ2.1 (`thesisrq15acr.azurecr.io/thesis-inference @sha256:d292aef407ee…`), recorded in `image_reference.txt`.

Table 5.26: RQ2.2 benchmark configuration. Sources: `rq22_rolling_summary.json`, `rq22_terraform.tfvars.json`, and `image_reference.txt` in `rq2_2_confidential_20260516_135726`.

Setting	Value
Topology	`chain_2svc`; split after `layer2`
Communication security	Managed-Istio mTLS, service identity, and authorisation policy inherited from RQ2.1
Standard VM size	`Standard_D8as_v5` (both services in standard condition; service1 in confidential condition)
Confidential VM size	`Standard_DC8as_v5` with AMD SEV-SNP (service2 in confidential condition)
TEE scope	`service2_only`; full two-service TEE is out of scope (future work)
Region	`westeurope`
Model	ResNet-18 (pretrained)
Device / precision	CPU / FP32
Input shape	$1 \times 3 \times 224 \times 224$
Paired passes	5
Warmup iterations	50 per condition execution
Measured iterations	200 per condition execution; 1,000 per condition after merging
Artifact	`rq2_2_confidential_20260516_135726`

The benchmark compared two conditions. In the standard condition, service1 and service2 both ran on `Standard_D8as_v5`. In the confidential condition, service1 remained on `Standard_D8as_v5`, while service2 ran on `Standard_DC8as_v5` backed by AMD SEV-SNP. This isolates the cost of protecting the downstream service that receives intermediate activations and executes the later ResNet-18 segment. The rolling design alternates standard and confidential passes from a seeded order (standard-first on passes 1, 3, 5; confidential-first on passes 2, 4 per `execution_plan.json`), and parity validation against the monolithic reference (`max_abs_diff` $= 0.0$, `num_inputs` $= 5$) passed in every per-pass `parity_validation.json`.

Table 5.27: RQ2.2 service2-only confidential execution results. Source: per-pass `benchmark/raw_iterations.csv` in `rq2_2_confidential_20260516_135726/conditions/`, aggregated across the five standard and five confidential passes (1,000 iterations per condition).

Condition	Service2 VM	n	Mean (ms)	Median (ms)	p95 (ms)
`standard`	`Standard_D8as_v5`	1000	58.492	58.283	60.255
`confidential`	`Standard_DC8as_v5`	1000	60.346	59.756	63.856
Delta	—	—	**+1.854 / +3.17%**	**+1.473 / +2.53%**	**+3.601 / +5.98%**

5.8.3 Results

In the service2-only TEE run, mean end-to-end latency increased from 58.492 ms to 60.346 ms. The mean overhead was therefore 1.854 ms, or 3.17%. Median latency increased by 1.473 ms, and p95 latency increased by 3.601 ms. Sequential throughput, computed from the mean latency, decreased from 17.097 req/s to 16.571 req/s, a reduction of approximately 3.08%.

This is the only confidential-execution scope evaluated in the thesis. The evidence manifest scopes RQ2.2 to `service2_only` and explicitly treats a full two-service confidential placement as future work. The trade-off table in section 5.9 therefore compares plain AKS, mTLS, and mTLS plus selective TEE only.

The interpretation of these results and the answer to RQ2.2 are presented in section 6.3.2.

5.9 RQ2.3 Security-Hardening Trade-off Synthesis

5.9.1 Research Question

This subsection addresses the following sub-research question:

> Which security-hardening configuration provides the most appropriate trade-off between performance cost, operational complexity, and protection scope for the selected AKS microservice inference deployment?

RQ2.3 is a synthesis question rather than a new benchmark. RQ2.1 measured the cost of adding service identity, mutual TLS, and inter-service authorization policy to the selected Azure inference path. RQ2.2 then measured the additional cost of placing the downstream protected service on an AMD SEV-SNP confidential VM node pool while keeping the communication-security contract fixed. RQ2.3 combines these results to compare the evaluated hardening levels and identify when each level is appropriate.

5.9.2 Synthesis Basis

The synthesis compares three deployment configurations evaluated in this thesis. The first is the plain AKS baseline from RQ1.5, where the inference chain runs without service-mesh mTLS or confidential-node placement. The second is the RQ2.1 communication-secured configuration, where managed-Istio mTLS, service identity, and authorization

policy are added to the selected chain. The third is the RQ2.2 selective confidential-execution configuration, where only the downstream protected service runs on an AMD SEV-SNP confidential node. Extending the confidential-computing boundary to both services (full two-service TEE) is out of the thesis-facing scope per the evidence manifest and is discussed only as future work.

These configurations should not be interpreted as interchangeable variants of the same security mechanism. They protect different parts of the system. Plain AKS provides the performance baseline but does not add the communication-level identity and encryption policy evaluated in RQ2.1. The mTLS/AuthZ configuration hardens the service-to-service path by authenticating workloads and enforcing authorization policy. The confidential-node configuration adds a VM-level runtime protection boundary for the downstream service placed on AMD SEV-SNP infrastructure. As discussed in sections 4.7.12, 6.3.2 and 6.6, the SEV-SNP threat-model scope is bounded: residual attack surfaces such as malicious-host interrupt injection remain outside the memory-encryption guarantee [29], and this thesis does not experimentally evaluate such attacks.

5.9.3 Trade-off Summary

The trade-off is not a single global ranking because the preferred configuration depends on the threat model. If performance is the only criterion, the plain AKS deployment is the best option. However, it does not provide the additional service identity, encrypted inter-service communication, or authorization-policy enforcement evaluated in RQ2.1. If the main security requirement is to protect communication between inference services, the mTLS/AuthZ configuration provides the most appropriate default balance. It adds a mean latency overhead of 3.488 ms, or 4.62%, for the selected `chain_2svc` topology, while enforcing the intended service-account-based policy. The `chain_5svc` stress case shows that this cost grows with the number of protected boundaries, reaching 8.837 ms, or 10.74%, when the full five-service chain is protected.

If the threat model includes host-level exposure of the downstream inference component, mTLS alone is not sufficient because it protects the communication path rather than the runtime memory of the service, and prior work shows that an untrusted remote service holding only intermediate feature maps can in principle recover the input [12]. Selective confidential execution provides a narrower hardening step against an infrastructure-level adversary on the host platform. The service2-only confidential-node run increases mean latency by 1.854 ms, or 3.17%, relative to its paired standard-node mTLS baseline. This makes selective TEE attractive when the downstream service is the main infrastructure-level sensitivity boundary, while bounded by SEV-SNP's known residual attack surfaces [29], which this thesis does not experimentally evaluate.

Extending the confidential-execution boundary to both services would broaden the runtime-protection scope at additional cost. That configuration is out of the thesis-facing scope of RQ2.2 (see the evidence manifest) and is therefore discussed as future work rather than as a measured trade-off point.

The interpretation of these results and the answer to RQ2.3 are presented in section 6.3.3.

Table 5.28: RQ2.3 synthesis of evaluated security-hardening levels. mTLS overheads are taken from table 5.22; the selective TEE overhead is taken from table 5.27 relative to its own paired standard-node mTLS baseline.

Configuration	Protection scope	Measured cost	Thesis interpretation
Plain AKS	Baseline AKS deployment without mesh mTLS/AuthZ or confidential-node placement.	Fastest evaluated Azure deployment path.	Best latency baseline, but weakest protection in the evaluated security set. Suitable only when additional communication and runtime hardening are not required.
mTLS/AuthZ	Service identity, encrypted service-to-service communication, and authorization policy for the selected inference path.	chain_2svc: +3.488 ms / +4.62%; chain_5svc stress case: +8.837 ms / +10.74%.	Best general-purpose hardening layer when the main requirement is authenticated and encrypted inter-service communication. Cost is measurable but bounded for the selected chain, and grows with protected chain depth.
mTLS/AuthZ + selective TEE	Communication hardening plus AMD SEV-SNP VM-level confidential execution for the downstream protected service (`service2_only`).	+1.854 ms / +3.17% over the newly paired standard-node mTLS baseline.	Adds runtime protection for the downstream segment that receives intermediate activations [12], bounded by the SEV-SNP threat model [29]. Full two-service TEE is treated as future work.

Chapter 6

Analysis

This chapter interprets the empirical findings reported in chapter 5 and answers the research questions defined in chapter 1. Sections 6.1 to 6.3 provide per-RQ interpretation grouped by research strand and close with a cross-RQ synthesis. Section 6.4 consolidates the evidence that supports treating the conclusions as credible. Section 6.5 situates the system against the alternatives it was compared with. Section 6.6 consolidates the threats to validity.

6.1 Architectural Split Choice

6.1.1 RQ1.1: Coarse Boundary Selection

The observed pattern in section 5.1 aligns with prior literature on split computing and distributed CNN inference. First, the poor performance of the earliest split is consistent with the widely reported effect that early feature maps are expensive to transmit because they remain large in spatial dimensions [31, 33]. Second, the late split after `layer4` shows that minimizing activation size alone is not sufficient; a split must also preserve a meaningful compute balance across services [20, 22]. Third, the favorable position of `split_after_layer2` and `split_after_layer3` is consistent with prior work that treats split placement as a trade-off between communication and computation rather than as a pure activation-size minimization problem [18, 20, 33].

The benchmark also indicates that raw latency ranking alone is not sufficient for the thesis-facing carry-forward decision. Under the predeclared selection rule, `split_after_layer4` is excluded from main-candidate eligibility because it is compute-degenerate, regardless of its absolute latency. Among the remaining non-degenerate candidates, `split_after_layer3` achieved the lowest raw split mean at 79.08 ms and `split_after_layer2` reached 80.12 ms (table 5.2). The frozen `carry_forward.json` in `rq1_1_20260515_111428` therefore selects `split_after_layer3` as the main carry-forward candidate and retains `split_after_layer2` as the reference candidate.

The results therefore support the thesis hypothesis that coarse architectural stage boundaries are not equally suitable for microservice-based inference. Rather, the most suitable boundaries are those that reduce activation-transfer burden without collapsing

the second partition into a near-empty remainder. In this experiment, the strongest balanced carry-forward region lies in the mid-to-late coarse portion of the network around `layer2` and `layer3`, while the latest split after `layer4` remains informative as a low-transfer, structurally degenerate boundary.

Answer to RQ1.1. RQ1.1 shows that coarse architectural stage boundaries are not equally suitable for two-part microservice decomposition of ResNet-18 under controlled local execution. Early boundaries incur high activation-transfer and boundary-crossing costs, while the latest boundary after `layer4` minimizes transfer size but becomes compute-degenerate. The most suitable balanced carry-forward candidates therefore lie in the mid-to-late coarse region around `layer2` and `layer3`, which provide the most credible trade-off between end-to-end latency, communication-related cost, and compute distribution. According to the screening logic, `split_after_layer3` is selected as the main coarse carry-forward candidate, with `split_after_layer2` retained as the reference candidate [18, 20, 33].

6.1.2 RQ1.2: Fine-Grained Refinement

The results in section 5.2 support three observations.

The refined region remains tightly clustered. All three refined split conditions fall within a very narrow latency band, from 80.16 ms to 80.39 ms (table 5.4). This indicates that the block-level refinement does not reveal a dramatically superior new split. Instead, it confirms that the carried-forward `layer2`–`layer3` region identified in RQ1.1 already captures the main structure of the latency–communication trade-off in this late-stage part of the network.

Equal activation size does not guarantee equal performance. The most instructive comparison in this refinement stage is between `split_after_layer3_block0` and `split_after_layer3`. Both conditions transfer an intermediate tensor of approximately 196 KB, yet they differ in boundary-crossing interval by approximately 7.8 ms and in mean end-to-end latency by a smaller but still measurable amount. This demonstrates that activation-transfer size is not the sole determinant of boundary cost. The distribution of computation on either side of the boundary also matters: shifting the split from the end of `layer3` to the internal boundary after `layer3.0` changes the compute placement and therefore changes boundary-crossing behavior even when the serialized tensor is the same size. This finding is consistent with the broader partitioning literature, which treats compute distribution as a first-class consideration alongside communication cost [18, 28, 31–33].

This run does not overturn the coarse screening. Under the current refined benchmark outcome, `split_after_layer3_block0` is the raw-fastest split and is selected as the main candidate by the implemented carry-forward rule in `rq1_2_20260515_112636/carry_forward.json`, with `split_after_layer2` retained as the reference candidate. However, the three refined conditions remain so close that the central thesis-facing conclusion is not that

RQ1.2 discovered a dramatically better split than RQ1.1. Rather, the refinement confirms that the late carried-forward region remains the correct focus, while revealing that small block-level shifts can still change behavior measurably even when activation-transfer size is held constant.

Answer to RQ1.2. RQ1.2 shows that finer-grained refinement within the carried-forward coarse region from RQ1.1 does not overturn the coarse-level findings, but it does reveal a meaningful secondary insight. In the refined comparison, `split_after_layer3_block0` is the raw-fastest split and is selected as the main candidate under the implemented selection rule, while `split_after_layer2` is retained as the reference candidate. The refined split `split_after_layer3` is slightly slower in this run. At the same time, the comparison between `split_after_layer3_block0` and `split_after_layer3` — which share the same activation-transfer size of approximately 196 KB yet differ in boundary-crossing interval by approximately 7.8 ms — demonstrates that equal activation-transfer size does not guarantee equal performance. Compute placement within the refined region still matters, even when the serialized tensor is held constant. The coarse RQ1.1 screening had already identified the correct late-stage comparison space; the block-level refinement confirms that structure and adds the observation that per-block compute distribution is a secondary but observable factor in boundary-crossing cost [15, 18, 21, 28, 31–33].

6.1.3 RQ1.3: Explanatory Synthesis

The explanatory analysis in section 5.3 supports a coherent two-dimensional account of the local findings from RQ1.1 and RQ1.2.

At the coarse level (RQ1.1), activation-transfer size is the dominant factor explaining the gradient of boundary-crossing costs from early to late splits (53.81 ms, 40.85 ms, 26.06 ms, and 2.22 ms after `layer1` through `layer4` respectively, per `rq1_1_20260515_111428/condition_summaries.csv`). The monotonic relationship between activation size and boundary-crossing interval is strong and consistent. However, this single dimension is insufficient: the compute-degenerate `split_after_layer4` minimises transfer burden but fails as a meaningful decomposition because it collapses one side of the partition into a near-empty service.

At the refined level (RQ1.2), compute distribution emerges as an independent explanatory dimension. The controlled comparison between two conditions with identical activation size but different boundary-crossing intervals shows that the placement of computation around the boundary affects cost even when communication burden is held constant. The internal timing experiment supports this observation by confirming that per-stage and per-block compute within ResNet-18 is structurally uneven, with `layer4` accounting for nearly a third of total forward-pass time and individual blocks within `layer3` contributing unequally. The operation-level support data strengthens this explanation further by showing that the heaviest internal units are overwhelmingly convolutional and concentrated in the later part of the network.

Neither dimension alone is sufficient. Activation-transfer size explains the broad coarse pattern but not the refined within-region differences. Compute distribution explains the refined differences but does not account for the large boundary-crossing cost gradient between early and late coarse splits. Together, they provide a more complete

explanation: suitable partition boundaries arise from a trade-off between activation-transfer burden, boundary-crossing cost, and compute distribution, not from the minimisation of any single metric.

Answer to RQ1.3. RQ1.3 shows that the latency and boundary-crossing behavior observed in RQ1.1 and RQ1.2 can be explained by two complementary structural properties of the partition: activation-transfer size and compute distribution.

Activation-transfer size is the dominant factor at the coarse level. The monotonic decrease in boundary-crossing interval from early to late splits in RQ1.1 closely tracks the reduction in intermediate tensor size as the boundary moves deeper into the network. This explains why early splits are costly and why later splits generally incur lower communication overhead.

Compute distribution is an independent and observable factor at the refined level. The RQ1.2 comparison between `split_after_layer3_block0` and `split_after_layer3` — which share the same activation size of approximately 196 KB yet differ in boundary-crossing interval by approximately 7.8 ms — demonstrates that equal transfer burden does not guarantee equal performance. The placement of computation around the boundary matters independently of the size of the transferred tensor.

The internal timing experiment supports the compute-distribution explanation by confirming that execution cost within ResNet-18 is structurally uneven: convolution operations account for approximately 82% of total compute, `layer4` alone accounts for 31.28%, and the heaviest internal units are concentrated in convolution-heavy late-stage regions (tables 5.6 and 5.7). These structural properties explain both the compute degeneracy of the `layer4` split observed in RQ1.1 and the boundary-crossing asymmetry between the two 196 KB conditions observed in RQ1.2.

Taken together, under these controlled local conditions, suitable microservice boundaries for ResNet-18 arise from a joint trade-off among activation-transfer burden, boundary-crossing cost, and compute distribution across services [7, 15, 18, 31]. No single metric is sufficient. The mid-to-late coarse region around `layer2` and `layer3` remains the strongest carry-forward zone because it avoids both the large activation costs of early boundaries and the poor compute balance of overly late boundaries. These local explanatory findings provide the interpretive basis against which the later Kubernetes and Azure stages can evaluate whether infrastructure context changes the relative importance of these two dimensions.

6.1.4 Synthesis: Architectural Split Choice

The local split-selection stages show that the position of the model boundary matters before Kubernetes or cloud effects are introduced. RQ1.1 found that coarse residual-stage boundaries are not interchangeable: early boundaries carry high activation-transfer cost, while the very late `layer4` boundary becomes compute-degenerate. The strongest carry-forward region is therefore the mid-to-late part of the network around `layer2` and `layer3`, as shown in table 5.2. This matches the broader split-computing literature, which treats partitioning as a trade-off among activation size, compute allocation, and deployment conditions [15, 18, 31, 33].

RQ1.2 refined this region and showed that a finer boundary can change behavior even when activation size is held constant. The comparison between `split_after_layer3_block0` and `split_after_layer3` is the important case: both transfer the same approximate activation size, but their boundary-crossing intervals differ. This supports the RQ1.3 interpretation that activation-transfer burden and compute placement are complementary explanations rather than competing ones [20, 28, 32].

6.2 Deployment Context

6.2.1 RQ1.4: Local Kubernetes Chain

The results in section 5.4 reveal four bounded findings for this controlled local Kubernetes benchmark.

Finding 1: Latency rises monotonically with service count. Mean end-to-end latency increases from 81.066 ms for `monolithic_k8s_1svc` to 83.387 ms, 87.728 ms, 90.572 ms, and 92.535 ms for `chain_2svc` through `chain_5svc`. Relative to the Kubernetes monolithic baseline, the added overhead grows from +2.321 ms (2.9%) to +11.469 ms (14.1%) (table 5.9). Under this controlled local Kubernetes setup, the evaluated coarse-stage ResNet-18 family shows a clear accumulation of cost as synchronous service boundaries are added.

Finding 2: Marginal cost shrinks for the latest boundary. The step from `chain_4svc` to `chain_5svc` increases mean latency by approximately 1.96 ms, smaller than the preceding 2.84 ms step from `chain_3svc` to `chain_4svc`. This is consistent with the recorded tensor-byte profile of the evaluated chain family: the frozen `total_activation_bytes_mean` metric rises from 1960 KiB to 2058 KiB, so the final condition adds the 98 KiB `layer4` output to the recorded per-request tensor total. In this setup, later boundaries can therefore add less overhead when the additional recorded tensor payload is small. This should be read as a property of the evaluated coarse-stage ResNet-18 chain family under this local benchmark, not as a universal law about microservice depth.

Finding 3: `chain_2svc` is the lowest-overhead non-monolithic condition. Among the chained deployments, `chain_2svc` remains the smallest non-monolithic overhead point at +2.321 ms (2.9%). Within the evaluated family, it is therefore the least expensive way to move from a single service to a chained deployment while preserving exact parity in the benchmarked workload. Because RQ1.4 uses a predefined condition set, this is an empirical summary of the final Kubernetes family rather than a carry-forward decision.

Finding 4: Scope of validity. Deployment metadata shows that all services ran on the same node of the local `kind` cluster. The result is therefore a valid local in-cluster Kubernetes benchmark, but it is still a single-node local cluster result. Broader cloud validation is deferred to RQ1.5.

Answer to RQ1.4. Under the final controlled local Kubernetes setup on the Ubuntu/Linux runtime, increasing the number of synchronously chained coarse-stage ResNet-18 services raises end-to-end latency from `monolithic_k8s_1svc` through `chain_5svc`, with mean latency increasing from 81.066 ms to 83.387 ms, 87.728 ms, 90.572 ms, and 92.535 ms. The smallest non-monolithic overhead is `chain_2svc` at +2.321 ms (2.9%). The final marginal increment from `chain_4svc` to `chain_5svc` is approximately 1.96 ms, smaller than the preceding 2.84 ms step. In this single-node local `kind` cluster, added boundaries therefore impose measurable but bounded overhead, and later boundaries can contribute less when the additional recorded tensor payload is small. Carry-forward is not applicable in this stage because the Kubernetes condition set was predefined rather than selected.

6.2.2 RQ1.5: AKS Transfer Validation

The results in section 5.5 reveal four bounded findings for the AKS transfer-validation stage.

Finding 1: AKS preserves the local Kubernetes condition ordering. The complete RQ1.4 ordering is preserved in AKS: the monolithic Kubernetes service is fastest, followed by `chain_2svc`, `chain_3svc`, `chain_4svc`, and `chain_5svc`. This provides directional transfer evidence. The same service-count trend that increased latency in local Kubernetes remains visible under managed cloud execution, even though the absolute millisecond values differ between environments.

Finding 2: AKS overhead is monotonic with service count. Relative to the AKS monolithic baseline, mean latency rises from 70.802 ms to 73.437 ms, 75.410 ms, 78.294 ms, and 80.519 ms as the chain grows from one to five services. The corresponding overheads are +2.635 ms (3.7%), +4.608 ms (6.5%), +7.492 ms (10.6%), and +9.717 ms (13.7%). Within this AKS deployment, adding synchronous service boundaries is therefore associated with measurable end-to-end cost.

Finding 3: Marginal overhead is bounded by the activation profile. The AKS marginal increments are +2.635 ms from the monolithic baseline to `chain_2svc`, +1.973 ms from `chain_2svc` to `chain_3svc`, +2.884 ms from `chain_3svc` to `chain_4svc`, and +2.225 ms from `chain_4svc` to `chain_5svc`. The first three steps reflect the cost of adding synchronous boundaries that move 392–784 KiB activations, while the final step is smaller than the preceding one because the layer-4 boundary moves only 98 KiB. This is consistent with the bounded non-linearity expected from the activation profile of the chain family: an additional service boundary does not imply a constant per-boundary penalty.

Finding 4: Absolute latency is environment-specific. All AKS means are lower than their local Kubernetes counterparts, including the monolithic baseline. The lower AKS absolute latencies should not be interpreted as evidence that AKS is generally faster than local Kubernetes. The local and AKS runs differ in host CPU, kernel, virtualisation layer,

Kubernetes implementation, container runtime context, CPU scheduling context, and networking path. RQ1.5 therefore uses within-environment ordering and normalised overhead as the primary comparison, not direct absolute latency. On that basis, AKS provides directional transfer evidence for the RQ1.4 ordering while preserving the claim scope: absolute latency is deployment-specific, and the AKS run should not be interpreted as a numerical replication of the local Kubernetes measurements.

Answer to RQ1.5. Under the controlled AKS transfer-validation setup, the local Kubernetes service-count pattern transfers directionally to managed cloud execution. The AKS run preserves the complete RQ1.4 ordering: `monolithic_k8s_1svc` remains fastest at 70.802 ms, followed by `chain_2svc` at 73.437 ms, `chain_3svc` at 75.410 ms, `chain_4svc` at 78.294 ms, and `chain_5svc` at 80.519 ms. Relative overhead grows monotonically from 3.7% for `chain_2svc` to 13.7% for `chain_5svc`, with the final marginal increment from `chain_4svc` to `chain_5svc` (+2.225 ms) smaller than the preceding step (+2.884 ms), consistent with the activation-transfer-size explanation from RQ1.3. RQ1.5 therefore supports a directional transfer claim: the same architectural chain ordering remains visible in AKS, although absolute latency values are environment-specific and should not be interpreted as a direct replication of the local Kubernetes measurements.

6.2.3 RQ1.5b: AKS Multi-Node Sensitivity

The results in section 5.6 support three bounded findings for the AKS multi-node sensitivity stage.

Finding 1: Multi-node placement preserves the AKS condition ordering. With every chain segment forced onto a distinct node and the benchmark client on a dedicated node, the complete RQ1.5 ordering is preserved (`monolithic_k8s_1svc` $<$ `chain_2svc` $<$ `chain_3svc` $<$ `chain_4svc` $<$ `chain_`
The bounded-non-linearity pattern from RQ1.4 and RQ1.5 also reproduces under multi-node execution: the `chain_4svc` $\rightarrow$ `chain_5svc` increment is 2.502 ms, broadly comparable to the preceding 4.562 ms step (table 5.18). The architectural ordering is therefore not an artefact of single-node colocation.

Finding 2: The inter-node penalty is positive and bounded. Table 5.19 compares the multi-node run condition-by-condition with the frozen single-node RQ1.5 baseline. The monolithic condition gains 2.306 ms of latency relative to the single-node monolithic, because the client now ships a 600 KiB input tensor across the Azure VNet rather than across a loopback socket. The chained conditions add per-condition deltas between 2.131 ms (`chain_3svc`) and 4.086 ms (`chain_5svc`). The multi-node penalty is therefore a measured bound rather than an open acknowledgement: the worst-case single-node-to-multi-node delta on the evaluated chain family is 4.086 ms (`chain_5svc`), and the monolithic baseline alone gains 2.306 ms.

Finding 3: Round-to-round consistency remains low. Round CVs for the chained conditions in RQ1.5b are 0.0036–0.0127, comparable to and in some cases tighter than the

single-node RQ1.5 chained CVs of 0.0019–0.0061 (tables 5.17 and 5.20). The monolithic condition shows a similar profile (0.0084 vs 0.0117). Cross-round stability is therefore preserved when the topology is distributed across nodes.

Answer to RQ1.5b. Under enforced multi-node placement, the same five-condition chain family preserves the monotonic overhead progression seen in RQ1.4 and RQ1.5: per-condition mean latency increases from 73.108 ms (monolithic) to 84.605 ms (`chain_5svc`), and the `chain_4svc` → `chain_5svc` step is +2.502 ms. The per-condition multi-node penalty relative to the single-node RQ1.5 baseline ranges from +2.131 ms to +4.086 ms across the chained conditions and +2.306 ms for the monolithic baseline. The single-node RQ1.4 / RQ1.5 design therefore underestimates true production latency by a bounded margin that is now quantified per condition rather than acknowledged abstractly.

6.2.4 Synthesis: Deployment Context

RQ1.4 and RQ1.5 show that the architectural pattern remains visible after the workload is moved into Kubernetes-style service chains. In local `kind`, latency increases from the monolithic Kubernetes baseline through the four-service chain, while the final five-service increment is small because the added late-stage boundary transfers a much smaller activation tensor. In AKS, the same condition ordering is preserved: `monolithic_k8s_1svc`, `chain_2svc`, `chain_3svc`, `chain_4svc`, and `chain_5svc`. The absolute millisecond values differ between local Kubernetes and AKS, but the within-environment overhead trend transfers directionally, as summarised in table 5.16.

This distinction is important. The thesis does not claim that local Kubernetes predicts AKS latency numerically. Cloud and Kubernetes performance studies show that container runtime, virtualization, CNI, managed-cluster implementation, and cloud variability can change absolute measurements [6, 8, 9, 16]. The stronger supported claim is architectural: when the same ResNet-18 chain family is run in AKS, the relative ordering remains interpretable even though the platform changes.

6.3 Security Hardening

6.3.1 RQ2.1: Service Identity and Mutual TLS

The results in section 5.7 support four bounded findings.

Finding 1: the mTLS/AuthZ hardening bundle adds measurable but bounded overhead to the selected chain. For the primary `chain_2svc` topology, the service-mesh hardening layer adds 3.488 ms, or 4.62%, to mean end-to-end latency (table 5.22). This is a measurable cost, but it is bounded relative to the approximately 75.6 ms plain AKS-with-mesh-tainted baseline. The result is consistent with service-mesh literature: sidecars add critical-path work, but the visible percentage overhead depends on the workload and on how much application compute the proxy cost is amortised over [36].

Finding 2: security overhead grows with protected chain depth. The `chain_5svc` stress run increases the mean hardening overhead to 8.837 ms (10.74%). This is approximately two and a half times the absolute `chain_2svc` overhead. The larger cost is expected: the stress topology contains more injected proxies, more authorization policies, and more protected inter-service boundaries. This finding reinforces the RQ1 result that service-count decisions and security decisions cannot be separated. A chain that is tolerable without communication hardening may become less attractive once every boundary also carries identity, encryption, proxying, and policy-enforcement cost.

Finding 3: the result is lower than some published Istio overheads, but not in conflict with them. The observed 4.62% primary overhead is much smaller than the high-load Istio penalties reported in broader service-mesh benchmarks [5]. This does not contradict that literature. Those studies use different services, protocols, load levels, tail-latency metrics, mesh configurations, and deployment environments. RQ2.1 uses a CPU-bound ResNet-18 inference chain with substantial per-request model computation, same-node placement, and a specific managed-Istio AKS configuration. The correct comparison is therefore qualitative: both this thesis and the literature show that sidecar-mediated mTLS has non-zero latency and resource cost, while the absolute magnitude is workload-specific.

Finding 4: RQ2.1 hardens inter-service communication, not the entire trust model. The security validation shows that the intended service-account-based communication policy was enforced for the protected downstream segments: valid full-chain requests succeeded, while direct non-mesh probes to protected downstream services were blocked. This is meaningful communication-level hardening. However, it does not remove trust in the service-mesh control plane, does not protect against all Kubernetes administrator threats, and does not provide confidential execution for model code or activations. Those limits matter because prior work identifies the mesh control plane and local CA as security-critical trust points [27]. RQ2.1 should therefore be claimed as a measured first hardening layer, not as a complete security solution.

Answer to RQ2.1. Under the paired AKS RQ2.1 benchmark, adding service identity, managed-Istio mTLS, and inter-service authorization policy introduces measurable but bounded overhead for the selected `chain_2svc` deployment. Mean latency increases from 75.557 ms in the plain condition to 79.045 ms with the mTLS/AuthZ hardening bundle, an overhead of 3.488 ms or 4.62%; p95 latency increases by 3.623 ms. Security validation passed in every mTLS pass, including full-chain positive probes and direct-denial probes against protected downstream segments. The additional `chain_5svc` stress run shows that the same hardening approach becomes more expensive as service depth increases: mean overhead rises to 8.837 ms or 10.74%. RQ2.1 therefore supports a cautious conclusion: service identity and managed-Istio mTLS/AuthZ are feasible for the selected Azure chain at a modest latency cost, but the cost is not negligible, grows with the number of protected service boundaries, and comes with operational complexity from sidecars, policies, resource requests, and readiness delay. The follow-up ablation

indicates that this cost is dominated by entering the managed-Istio sidecar data path rather than by strict mTLS enforcement or AuthorizationPolicy evaluation alone.

6.3.2 RQ2.2: Confidential VM Execution

The results in section 5.8 support three bounded findings.

Finding 1: VM-level confidential execution adds measurable overhead on top of mTLS. The RQ2.2 paired run shows positive confidential-execution overhead relative to a newly collected standard baseline. This matters because the baseline already includes the RQ2.1 mTLS/AuthZ hardening. The measured delta therefore represents the additional cost of placing the downstream service on confidential VM infrastructure, not the cost of introducing service-mesh communication security.

Finding 2: selective TEE has a modest measured cost on the protected segment. Placing only `service2` on `Standard_DC8as_v5` (AMD SEV-SNP) increases mean latency by 1.854 ms (+3.17%), median latency by 1.473 ms (+2.53%), and p95 latency by 3.601 ms (+5.98%) relative to the matched standard-node baseline (table 5.27). This is the most direct implementation of the original RQ2.2 design: `service2` is the downstream protected component, receives intermediate activations from `service1`, and is the service guarded by the strict mTLS and authorisation policy in the RQ2.1 contract. He et al. [12] motivate protecting that intermediate activation beyond the in-transit mTLS layer, although they target a malicious-remote-service threat model that differs from the infrastructure-level adversary SEV-SNP defends against; selectively placing `service2` on confidential infrastructure therefore gives the thesis a narrow, defensible hardening point with a modest measured cost.

Finding 3: the security claim is stronger than RQ2.1 but still bounded. RQ2.1 protected the service-to-service communication path. RQ2.2 adds a VM-level runtime protection boundary against host-level access for the confidentially placed service. This is a stronger security posture, but it is not complete application-level isolation. The guest OS, application process, model runtime, and local sidecar remain trusted parts of the confidential VM environment, and plaintext exists inside the endpoint boundary after mTLS termination. Recent work additionally shows that confidential VMs retain residual attack surfaces beyond memory encryption, including malicious interrupt injection from the untrusted hypervisor [29]; this thesis does not experimentally evaluate such attacks. The correct claim is therefore that RQ2.2 adds VM-level confidential execution to the already communication-secured Azure inference chain, not that it removes every trusted component or eliminates all possible data exposure.

Answer to RQ2.2. RQ2.2 shows that AMD SEV-SNP VM-level confidential execution is feasible for the communication-secured AKS inference chain and introduces a measurable but bounded additional overhead. With only the protected downstream service moved to a confidential node (`service2_only` scope), mean latency increases by 1.854 ms, or 3.17%, and p95 latency increases by 3.601 ms, relative to a newly paired

standard-node mTLS baseline. Extending the confidential-execution boundary to both services in the chain (full two-service TEE) is out of the thesis-facing scope of RQ2.2 per the evidence manifest and is treated as future work. The interpretation remains bounded by the SEV-SNP threat model and by the sidecar-based service-mesh architecture: RQ2.2 evaluates VM-level confidential execution on Azure for the downstream segment, not application-level enclave isolation or complete end-to-end secrecy.

6.3.3 RQ2.3: Trade-off Synthesis

Section 5.9 shows that the evaluated security mechanisms form a cumulative hardening gradient rather than a single binary secure/insecure choice. The RQ1.5 plain AKS deployment establishes the performance baseline. RQ2.1 adds communication-level security through mTLS, service identity, and authorization policy. RQ2.2 then adds VM-level confidential execution for the downstream protected service (`service2_only`). Each step increases the protection scope, but each step also introduces additional latency, resource, or operational cost. Extending the confidential-execution boundary to both services (full two-service TEE) is treated as future work and is not evaluated as a measured trade-off point in this thesis.

The main engineering implication is that security hardening should be scoped to the threat model. Applying every available mechanism everywhere is not always the best default. In the evaluated deployment, mTLS/AuthZ is the most suitable general-purpose security baseline because it protects the inter-service path at a modest cost for `chain_2svc`. Selective TEE on the downstream service is the next available hardening step when that segment is the primary sensitivity boundary [12], while its security claim remains bounded by the SEV-SNP threat model [29].

The results also show that performance and security cannot be evaluated independently. RQ1 established that service boundaries add measurable overhead and that this overhead depends on boundary placement and service count. RQ2 shows that security mechanisms compound this cost. The mTLS/AuthZ hardening overhead grows when more service boundaries are protected, and selective confidential execution adds further overhead on top. The final deployment decision therefore depends on three linked factors: the chosen model partition, the number of service boundaries, and the required protection scope.

Answer to RQ2.3. RQ2.3 shows that the best hardening level depends on the assumed threat model and the acceptable performance cost. Plain AKS is the fastest evaluated configuration, but it provides the weakest protection in the security comparison. mTLS with service identity and authorization policy is the best general-purpose balance when the main goal is authenticated and encrypted service-to-service communication; in the selected `chain_2svc` deployment it adds 3.488 ms, or 4.62%, mean latency overhead. Selective TEE on `service2_only` is the preferred option evaluated in this thesis when the downstream inference segment is the main infrastructure-level sensitivity boundary, adding 1.854 ms, or 3.17%, over its paired standard-node mTLS baseline. Extending the confidential-execution boundary to both services (full two-service TEE) is treated as future work and is not evaluated here as a measured trade-off point.

Overall, the evaluated configurations support a cumulative security-performance trade-off: stronger protection is feasible for the selected AKS inference chain, but each additional protection layer adds measurable cost. The most defensible default is mTLS/AuthZ for communication-level protection, with selective confidential execution on the downstream service added only when the threat model requires runtime protection against host-level exposure, and bounded by the SEV-SNP attack-surface limitations discussed above [29].

6.3.4 Synthesis: Security Hardening

The security stages show that hardening is feasible for the selected AKS deployment, but it is not free. RQ2.1 adds managed-Istio mTLS, service identity, and authorization policy. For the selected `chain_2svc` topology, the measured mean overhead is 3.488 ms, or 4.62%; for the `chain_5svc` stress topology, the overhead grows to 8.837 ms, or 10.74% (tables 5.22 and 5.23). This is consistent with service-mesh studies showing that sidecar-based mTLS adds latency and resource cost through additional critical-path processing [5, 36].

RQ2.2 adds VM-level confidential execution on an AMD SEV-SNP node for the downstream protected service while keeping the mTLS contract fixed. Selective confidential execution for `service2_only` adds 1.854 ms, or 3.17%, over its paired standard-node mTLS baseline. The result is consistent with the confidential-computing literature's expectation that CVM overhead is workload- and scope-dependent [24, 25, 34], and the security claim is bounded by SEV-SNP's residual attack surfaces [29].

6.4 Validation

This section consolidates the evidence that supports treating the thesis conclusions as credible. The validation discipline follows the shared benchmark contract defined in sections 4.3 to 4.5.

Functional parity. Every thesis-facing frozen run achieves `max_abs_diff` $= 0.0$ against the monolithic reference under absolute tolerance 1.0×10^{-5} and five validation inputs. This applies to all five RQ1.1 conditions (section 5.1), all four RQ1.2 conditions (section 5.2), all five RQ1.4 conditions (section 5.4), all five RQ1.5 conditions (section 5.5), all five RQ1.5b multi-node conditions (section 5.6), and all paired RQ2.1 and RQ2.2 runs (sections 5.7 and 5.8). No numerical drift is introduced by the partition or transport layer at any stage.

Cross-round consistency. Each benchmark stage reports per-condition round CVs computed as the standard deviation of per-round condition means divided by the grand mean (section 4.5). The local screening stages (RQ1.1, RQ1.2) achieve round CVs in the range 0.001–0.015. The local Kubernetes stage (RQ1.4) sees round CVs of 0.0074–0.0161 (table 5.12). The AKS single-node stage (RQ1.5) sees round CVs of 0.0075–0.0225 across the chained conditions and the monolithic baseline (table 5.17); the AKS multi-node sensitivity stage (RQ1.5b) tightens these to 0.0020–0.0099 (table 5.20) because each

pod has its own dedicated VM and therefore no neighbour contention. These low CVs support treating the reported condition means as representative of each environment rather than as artefacts of a single favourable or unfavourable run.

Paired execution and transfer validation. RQ2.1 uses a paired alternated design (three mTLS-first passes, two plain-first passes) to defend against slow performance drift over the course of the benchmark. Pass-level mTLS deltas remain positive in every pass: 2.693–3.318 ms for `chain_2svc` and 8.203–9.968 ms for `chain_5svc` (table 5.23). RQ1.5 preserves the complete RQ1.4 condition ordering (`monolithic_k8s_1svc` $<$ `chain_2svc` $<$ `chain_3svc` $<$ `chain_4svc` $<$ `chain_5svc`) under managed-AKS execution (table 5.16). This directional transfer supports treating the chain-cost trend as architectural rather than as an environment-specific artefact, in line with systems-benchmarking guidance [13].

Security validation. The RQ2.1 mTLS configuration was validated through static manifest checks of peer-authentication and authorisation-policy objects, runtime validation of policy object counts, full-chain positive probes confirming that valid upstream-to-downstream requests succeeded, and direct non-mesh denial probes confirming that unauthenticated access to protected downstream services was blocked (section 5.7). All four checks passed in every paired mTLS pass. RQ2.2 collected fresh paired standard baselines for each confidential-node artifact rather than reusing older RQ2.1 data (section 5.8); incremental overhead can therefore be attributed to the confidential-node placement rather than to session-level drift.

Effect-size discipline. Effect sizes are computed from pooled per-iteration data ($N = 1{,}000$ per condition). At this N, Mann–Whitney p-values are near-zero for any non-trivial difference and are reported only as supplementary evidence. Cohen's d and cross-round stability are treated as the more informative measures of effect magnitude (section 4.5 and table 5.11).

6.5 Evaluation

The main engineering lesson is that partitioning, deployment, and hardening cannot be optimised independently. The split point determines activation-transfer size and compute placement. Kubernetes and AKS add platform-specific communication and scheduling effects. Security hardening then compounds the cost of the selected service topology. This means that a decomposition that is acceptable without mTLS or confidential execution may become less attractive once every service boundary is protected.

For the evaluated workload, `chain_2svc` is the most defensible non-monolithic deployment point. It avoids the excessive boundary count of deeper chains, preserves a meaningful architectural split, and remains the lowest-overhead chained condition in both local Kubernetes and AKS. It also provides the practical base for RQ2: the mTLS/AuthZ hardening bundle is measurable but bounded on this topology, and selective TEE can protect the downstream service at a lower cost than full-chain confidential execution.

Across the evaluated stack, the monolithic Kubernetes baseline remains the fastest path; chained deployments pay an overhead that is monotonic with service count but bounded and non-linear; the local-to-AKS transfer is directional rather than numerical; mTLS adds a workload-amortised but measurable cost that grows with protected chain depth; and selective confidential-VM execution on the downstream service adds a further bounded cost on top of the communication-secured chain. Extending the confidential-execution boundary to both services is treated as future work.

6.6 Threats to Validity

The strongest validity support is internal consistency. The experiments use fixed model weights, fixed input shape, functional parity validation, repeated rounds, and a shared benchmark contract. The local and Kubernetes stages also use CPU-affinity and single-threading controls where the runtime permits. These choices follow systems benchmarking guidance that stresses controlled comparisons, explicit measurement context, and caution when interpreting performance numbers [13].

The main limitation is external validity. The remaining threats below are consolidated from the methodological scoping in chapter 4 and apply to all stages of the staged pipeline.

Single-node clusters. The thesis-facing RQ1.4 `kind` and RQ1.5 AKS primary runs are both single-node. Single-node deployments eliminate inter-node network hops and do not capture the scheduling and networking variability of a multi-node production cluster. RQ1.5b (section 5.6, section 6.2.3) addresses this directly with a multi-node AKS sensitivity stage that forces every chain hop across a node boundary; the measured per-condition multi-node penalty is bounded between +2.131 ms (`chain_3svc`) and +4.086 ms (`chain_5svc`), with the monolithic baseline gaining +2.306 ms (table 5.19). The single-node primary numbers should therefore be interpreted as a clean architectural-overhead measurement that excludes the inter-node network path; the multi-node sensitivity in section 5.6 provides a quantitative bound on what production multi-node deployments add on top.

CPU-only inference. All stages use CPU inference with a single thread. GPU-accelerated inference would substantially shift the compute distribution, reduce per-inference time, and change the relative weight of the gRPC boundary cost. The results are not directly generalisable to GPU-hosted microservice deployments.

Single model and workload. The thesis evaluates one model (ResNet-18) at batch size one with one input shape. Different architectures, larger activation maps, or higher-batch workloads may produce different cost curves and different optimal split regions.

Controlled conditions versus production variability. The local stages apply strict CPU stabilisation and achieve low cross-round variance. The Kubernetes and AKS stages are less controllable and exhibit higher variability. Neither environment captures the

full noise of a shared multi-tenant cloud cluster, where CPU steal, network congestion, and noisy neighbours can substantially affect tail latency.

Activation-protection motivation. He et al. [12] demonstrate that an untrusted remote service holding only intermediate feature maps can reconstruct the client's input, so in-transit encryption alone does not protect activations at rest on the downstream segment. This motivates evaluating the infrastructure-level protection of the downstream service (RQ2.2) in addition to communication-level hardening (RQ2.1). The two layers address different threat models: SEV-SNP defends against the cloud host, not against an actively malicious application running inside the protected VM, so it is not a direct solution to the malicious remote-service model in He et al.

mTLS threat-model scope. Sidecar-based mTLS protects the inter-proxy communication path but does not prevent plaintext from existing inside the endpoint after TLS termination. The security claims for RQ2.1 are explicitly scoped to communication-level hardening.

SEV-SNP threat-model scope. AMD SEV-SNP protects the VM boundary against the host platform but not against a compromised guest OS or application process, and several side-channel and availability threats fall outside its core guarantee. In particular, recent work shows that confidential VMs remain vulnerable to malicious-interrupt injection from the untrusted hypervisor [29]; this thesis does not experimentally evaluate such attacks. RQ2.2 therefore claims VM-level confidential execution for the downstream segment (`service2_only`), not complete application-level isolation. Extending the confidential boundary to both services in the chain is treated as future work.

Generalisation of the cited literature. Multi-node clusters, concurrent clients, GPUs, larger models, feature compression, autoscaling, and different service-mesh configurations could change the absolute overheads and possibly the preferred topology. The thesis therefore supports bounded claims about the evaluated system rather than a universal rule for all microservice inference deployments.

Chapter 7

Conclusion

7.1 Summary & Findings

This thesis evaluated ResNet-18 inference as a staged microservice system. The study started with controlled local split selection, moved the selected chain family into local Kubernetes and AKS, and then measured the additional cost of service-mesh mTLS, authorization policy, and AMD SEV-SNP confidential VM execution.

The main finding is that microservice inference cost is shaped by three linked factors: where the neural network is split, where the resulting services are deployed, and how strongly the communication and execution path are hardened. RQ1 showed that suitable ResNet-18 boundaries lie in the mid-to-late region around `layer2` and `layer3`, because early boundaries transfer larger activations and overly late boundaries become compute-degenerate. RQ1.4 and RQ1.5 then showed that the same service-count ordering remains visible in Kubernetes and AKS, although absolute latency is environment-specific. RQ1.5b additionally bounded the inter-node penalty that the single-node primary design intentionally hides: forcing every chain hop across a node boundary on a multi-node AKS cluster adds between +2.131 ms and +4.086 ms per chained condition, with a +2.306 ms baseline penalty on the monolithic configuration and the architectural ordering preserved. RQ2 showed that mTLS/AuthZ and selective AMD SEV-SNP confidential VM execution on the downstream service are feasible for the selected AKS chain, but each protection layer adds measurable overhead; the security claims are bounded by the mTLS termination boundary and by the SEV-SNP threat model [12, 29].

7.2 Contributions

The thesis makes three contributions. First, it provides a staged empirical evaluation method for neural-network microservice decomposition, moving from local split selection to Kubernetes, AKS, and security hardening without changing the model or benchmark contract. Second, it shows that activation-transfer size and compute placement must be considered together when selecting ResNet-18 service boundaries. Third, it quantifies a security-performance trade-off for the selected Azure deployment path: mTLS/AuthZ is a practical communication-hardening baseline, and selective AMD SEV-

SNP TEE on the downstream service adds an additional bounded cost while leaving the SEV-SNP residual threat model unchanged. Extending the confidential-execution boundary to both services in the chain is treated as future work rather than as a measured trade-off point.

7.3 Limitations & Threats to Validity

The conclusions are bounded by the evaluated workload and deployment setup (ResNet-18, CPU-only FP32 inference, batch size one, synchronous gRPC, and same-node Kubernetes and AKS placement for the thesis-facing primary runs) and by the bounded threat models of the evaluated security mechanisms. The RQ1.5b sensitivity stage explicitly relaxes the same-node placement constraint to bound the inter-node network penalty, but other production factors (multi-region placement, autoscaling, concurrent client load, non-default CNI plugins) remain out of scope. The full discussion of these threats is consolidated in section 6.6.

7.4 Future Work

Future work should extend the evaluation to larger and more diverse models, including transformer-based inference workloads and models with different activation profiles. The same staged method could also be applied under concurrent load, GPU execution, cross-availability-zone or cross-region placement, and autoscaling conditions; the multi-node sensitivity stage in section 5.6 bounds the intra-zone inter-node penalty but leaves these wider topology questions open. Another useful extension is to combine architectural partitioning with feature compression, since compression may change the cost of early split points. Finally, the security evaluation could be expanded to compare sidecar and sidecarless meshes, application-layer encryption, attestation workflow integration, and different confidential-computing platforms. In particular, extending the confidential-execution boundary from the selective `service2_only` scope evaluated here to both services in the chain (full two-service TEE), and evaluating SEV-SNP residual attack-surface attacks such as malicious-host interrupts [29], are explicit next steps that fall outside the canonical evidence base of this thesis.

Appendix A

Appendix

This appendix collects supporting material that is auxiliary to the main argument: extended sensitivity data, additional ablation breakdowns, and any figures or tables that are too large for the experiment chapters. The thesis-facing primary results remain those reported in chapters 5 and 6.

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